Further Functions

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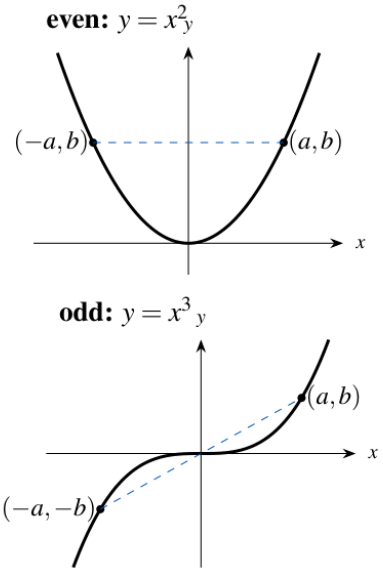
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1. Definitions and symmetry

Even: $f(-x) = f(x)$ for all x in the domain.
Graph is **symmetric about the y-axis**.
(a,b) on the graph $\Rightarrow$ $(-a,b)$ on the graph.

Odd: $f(-x) = -f(x)$ for all x in the domain.
Graph has **rotational symmetry** $(180°)$ **about the origin**.
(a,b) on the graph $\Rightarrow$ $(-a,-b)$ on the graph.
If 0 is in the domain, an odd function has $f(0) = 0$.



2. How to test a function

1. Replace every x by $(-x)$ and simplify $f(-x)$.
2. Compare: $f(-x) = f(x) \Rightarrow$ **even**; $f(-x) = -f(x) \Rightarrow$ **odd**; otherwise **neither**.
3. Check the **domain** is symmetric: $f(x) = \sqrt{x}$ is neither (no $f(-1)$).

Polynomials: only even powers (constant counts as x^0) $\Rightarrow$ even; only odd powers $\Rightarrow$ odd. $4x^3$ is odd — the coefficient does not matter.
Trig: cos is even; sin and tan are odd.

3. Combining functions

even $\pm$ even $=$ even	odd $\pm$ odd $=$ odd	even $\times$ even $=$ even
odd $\times$ odd $=$ **even**	even $\times$ odd $=$ odd	f odd $\Rightarrow$ $\lvert f\rvert$, f^2 even, $\frac{1}{f}$ odd

$f \circ g$: f even, g odd $\Rightarrow$ even $f \circ f$ with f odd $\Rightarrow$ odd $f(|x|)$ is **always** even

Proof style: "Let $h(x) = f(x)g(x)$. Then $h(-x) = f(-x)g(-x) = \cdots = h(x)$, so h is even."

Every f = even part $\dfrac{f(x)+f(-x)}{2}$ + odd part $\dfrac{f(x)-f(-x)}{2}$.

4. The modulus (absolute value)

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \qquad |x| = \sqrt{x^2} \qquad |x| = \text{distance from 0, so } |x| \geq 0.$$

Properties: $|-x| = |x|$, $|x|^2 = x^2$, $|xy| = |x||y|$, $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$, $|x-y| = |y-x|$.

Removing modulus signs: split at the point where the inside is 0, e.g. $x - 2|x| = \begin{cases} -x, & x \geq 0 \\ 3x, & x < 0 \end{cases}$

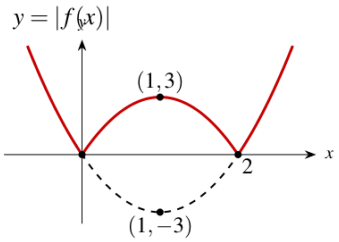
5. $y = |f(x)|$ and $y = f(|x|)$ (example: $f(x) = 3x(x-2)$)

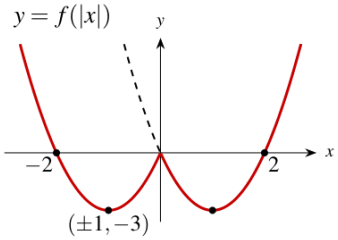
$y = |f(x)|$ (modulus *outside*):

- keep the graph where $f(x) \geq 0$;
- **reflect in the x-axis** where $f(x) < 0$;
- graph is never below the x-axis;
- points on the x-axis are invariant (sharp corners there).

$y = f(|x|)$ (modulus *inside*):

- discard the graph for $x < 0$;
- **reflect the part for $x \geq 0$ in the y-axis**;
- result is always an even function;
- points on the y-axis are invariant.





6. Effect on features

	$\lvert f(x)\rvert$	$f(\lvert x\rvert)$
x-intercepts	same	only those with $x \geq 0$, mirrored ($\pm$)
y-intercept	$\lvert f(0)\rvert$	same, $f(0)$
asymptotes	V.A. same; H.A. $y = k \to y = \lvert k\rvert$	V.A. $x = c > 0$ gives $x = \pm c$
domain / range	same domain; range ≥ 0	domain: $\lvert x\rvert$ in domain of f; range: cannot always tell

Both together $|f(|x|)|$: do $f(|x|)$ first (mirror right half), then reflect anything below the x-axis up.

7. Equations

- $|A| = k, k > 0 \Rightarrow A = \pm k$. $|A| = 0 \Rightarrow A = 0$. $|A|$ = negative $\Rightarrow$ **no solution**.
- $|A| = |B| \Rightarrow A = \pm B$ (or square both sides: $A^2 = B^2$).
- $\left|\dfrac{3x+2}{1-x}\right| = 4 \Rightarrow \dfrac{3x+2}{1-x} = \pm 4$, then multiply out each case.
- x **outside the modulus** (e.g. $|x+3| = 7 - \frac{x}{3}$): split into cases $x \geq -3$ and $x < -3$ and **reject** answers outside each case. Always check by substituting back.

8. Inequalities

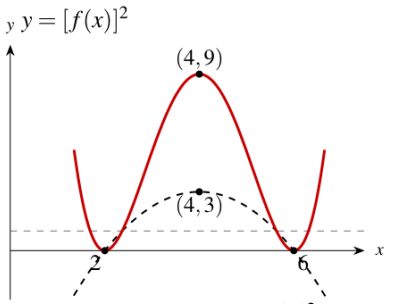
- $|A| < k \Leftrightarrow -k < A < k$ e.g. $|2x-1| < 3 \Rightarrow -1 < x < 2$
- $|A| > k \Leftrightarrow A < -k$ or $A > k$ e.g. $|4-x| \geq 5 \Rightarrow x \leq -1$ or $x \geq 9$
- $|A| < |B| \Leftrightarrow A^2 < B^2$ (both sides ≥ 0) $\Rightarrow$ quadratic inequality $\Rightarrow$ **sign diagram**.
- Fractions: $\left|\dfrac{x}{x+1}\right| \geq \dfrac{1}{2} \Rightarrow 4x^2 \geq (x+1)^2$, provided $x \neq -1$ (multiply by the *squared* denominator, which is positive).
- Never multiply an inequality by an expression unless you know its sign.

9. Graphical method (calculator questions) — TI-Nspire CX

1. home → **Add Graphs**; type `f1(x)=abs(2x-3)` and `f2(x)=x`. (`abs(` can be typed, or use the template palette.)
2. menu → **4: Window/Zoom** — start wide to see every crossing.
3. menu → **6: Analyze Graph** → **4: Intersection**; click to the left, then the right of each crossing.
4. Inequality: read off the x-values where f1 is *below* (for $<$) or *above* (for $>$) f2. Check the domain of each function (ln, $\sqrt{}$, arccos).

10. Sketching $y = [f(x)]^2$ from $y = f(x)$

- Points with $y = 0$ or $y = 1$ are **invariant**.
- The graph **touches** the x-axis at each x-intercept of f (new minimum).
- $[f(x)]^2 \geq 0$: graph is on or above the x-axis.
- Vertical asymptotes stay the same; H.A. $y = k$ becomes $y = k^2$.
- $-1 < f(x) < 1$: new graph is **closer** to the x-axis.
- $|f(x)| > 1$: new graph is **further** from the x-axis.
- Turning point (a, b) becomes (a, b^2).



f cuts at 2 and 6, max $(4,3)$; $[f]^2$ touches at 2 and 6, max $(4,9)$.

11. Domain, range and transformations

- Domain unchanged. Range: square the values; if the range of f includes 0, the new range starts at 0 (e.g. $-4 \leq y \leq 3 \;\rightarrow\; 0 \leq y \leq 16$).
- f odd $\Rightarrow [f(x)]^2$ even.
- $f(x) = 3^x$: $[f(x)]^2 = 3^{2x} = f(2x)$ — a horizontal stretch, scale factor $\frac{1}{2}$.
- $(g(x))^2 = r$: count intersections with the horizontal line $y = r$ (exam: "exactly two solutions").

12. Summary of the three graphs and TI-Nspire check

	$\lvert f(x)\rvert$	$f(\lvert x\rvert)$	$[f(x)]^2$
reflects	below x-axis $\rightarrow$ up	right half $\rightarrow$ left	— (squares y)
invariant	points with $y \geq 0$	points with $x \geq 0$	points with $y = 0$ or 1
at x-intercepts	sharp corner	mirrored	touches (smooth)

TI-Nspire: `f1(x)=...`, then `f2(x)=abs(f1(x))`, `f3(x)=f1(abs(x))`, `f4(x)=(f1(x))^2` to see all three at once.

FF1 Further Functions · original Q1 · source p.8

1

No calculator **Haese AA HL Ex 6A · Q1**

[Maximum mark: 2]

Show that $f(x) = \frac{1}{x^2} + 2$ is an even function.

FF2 Further Functions · original Q2 · source p.8

2

No calculator **Haese AA HL Ex 6A · Q2**

[Maximum mark: 2]

Show that $f(x) = x^3 - 3x$ is an odd function.

FF3 Further Functions · original Q3 · source p.8

3

No calculator **Haese AA HL Ex 6A · Q3**

[Maximum mark: 2]

Explain why $f(x) = 4x^3$ is an odd function, even though 4 is an even number.

FF4 Further Functions · original Q4 · source p.8

4

No calculator **Haese AA HL Ex 6A · Q4**

[Maximum mark: 6]

Determine whether each function is even, odd, or neither:

(a) $f(x) = 5x$ **(b)** $f(x) = -4x + 3$ **(c)** $f(x) = \frac{3}{x^2 - 4}$

(d) $f(x) = 2x^3 - \frac{5}{x}$ **(e)** $f(x) = x^2 + \frac{7}{x^2} - 3$ **(f)** $f(x) = \sqrt{x}$

FF1 Further Functions · original Q1 · source p.8

1 *Haese AA HL Ex 6A · Q1*

$f(-x) = \dfrac{1}{(-x)^2} + 2 = \dfrac{1}{x^2} + 2$ **M1**

$= f(x)$ for all $x \neq 0$, so f is **even**. **A1**

FF2 Further Functions · original Q2 · source p.8

2 *Haese AA HL Ex 6A · Q2*

$f(-x) = (-x)^3 - 3(-x) = -x^3 + 3x$ **M1**

$= -(x^3 - 3x) = -f(x)$, so f is **odd**. **A1**

FF3 Further Functions · original Q3 · source p.8

3 *Haese AA HL Ex 6A · Q3*

$f(-x) = 4(-x)^3 = -4x^3 = -f(x)$, so f is odd. **M1**

Odd/even depends on the **power of** x (here 3, odd), not on the coefficient. The constant 4 is unchanged when $x \to -x$. **R1**

FF4 Further Functions · original Q4 · source p.8

4 *Haese AA HL Ex 6A · Q4*

(a) $f(-x) = -5x = -f(x)$: **odd**

(b) $f(-x) = 4x + 3$, which is neither $f(x)$ nor $-f(x)$: **neither**

(c) $f(-x) = \dfrac{3}{x^2 - 4} = f(x)$: **even**

(d) $f(-x) = -2x^3 + \dfrac{5}{x} = -f(x)$: **odd**

(e) $f(-x) = x^2 + \dfrac{7}{x^2} - 3 = f(x)$: **even**

(f) domain $x \geq 0$ is not symmetric about 0 ($f(-1)$ is undefined): **neither**

FF5 Further Functions · original Q5 · source p.10

5

No calculator **Haese AA HL Ex 6A · Q5**

[Maximum mark: 2]

Suppose $f(x)$ is an even function, and $(1, 3)$ and $(-5, -2)$ lie on the graph of $y = f(x)$. Write down two other points which lie on the graph of $y = f(x)$.

FF6 Further Functions · original Q6 · source p.10

6

No calculator **Haese AA HL Ex 6A · Q6**

[Maximum mark: 2]

Suppose $g(x)$ is an odd function, and $(4, 6)$ and $(-1, 2)$ lie on the graph of $y = g(x)$. Write down two other points which lie on the graph of $y = g(x)$.

FF7 Further Functions · original Q7 · source p.10

7

No calculator **Haese AA HL Ex 6A · Q7**

[Maximum mark: 3]

Suppose $f(x) = (2x+3)(x+a)$, $a \in \mathbb{R}$ is an even function. Find the value of a.

FF8 Further Functions · original Q8 · source p.10

8

No calculator **Haese AA HL Ex 6A · Q8**

[Maximum mark: 3]

Suppose $g(x) = (x+1)\left(\frac{1}{x}+b\right)$, $b \in \mathbb{R}$ is an odd function. Find the value of b.

FF5 Further Functions · original Q5 · source p.10

5 *Haese AA HL Ex 6A · Q5*

Even: $(a,b) \to (-a,b)$ (reflection in the y-axis).

$\boxed{(-1,\,3)}$ and $\boxed{(5,\,-2)}$ **A1 A1**

FF6 Further Functions · original Q6 · source p.10

6 *Haese AA HL Ex 6A · Q6*

Odd: $(a,b) \to (-a,-b)$ (rotation of 180° about O).

$\boxed{(-4,\,-6)}$ and $\boxed{(1,\,-2)}$ **A1 A1**

FF7 Further Functions · original Q7 · source p.10

7 *Haese AA HL Ex 6A · Q7*

$f(x) = 2x^2 + (2a+3)x + 3a$ **M1**

even $\Rightarrow$ no odd power of x: $2a+3=0$ **M1**

$\boxed{a = -\dfrac{3}{2}}$ **A1**

FF8 Further Functions · original Q8 · source p.10

8 *Haese AA HL Ex 6A · Q8*

$g(x) = 1 + bx + \dfrac{1}{x} + b = bx + \dfrac{1}{x} + (1+b)$ **M1**

odd $\Rightarrow$ constant term must be 0: $1+b=0$ **M1**

$\boxed{b=-1}$ (then $g(x) = \dfrac{1}{x} - x$, which is odd) **A1**

FF9 Further Functions · original Q9 · source p.12

9

No calculator **Haese AA HL Ex 6A · Q9**

[Maximum mark: 6]

(a) If the quadratic function $f(x) = ax^2 + bx + c$, $a \neq 0$ is an even function, prove that $b = 0$.

(b) Under what conditions is the cubic function $g(x) = ax^3 + bx^2 + cx + d$, $a \neq 0$, an odd function?

(c) Under what conditions is the quartic function $h(x) = ax^4 + bx^3 + cx^2 + dx + e$, $a \neq 0$, an even function?

FF10 Further Functions · original Q10 · source p.12

10

No calculator **Haese AA HL Ex 6A · Q10**

[Maximum mark: 3]

Decide whether each of the following is even, odd, or neither. Explain how this is observed in the graph of the function.

(a) $f(x) = \cos x$ **(b)** $f(x) = \sin x$ **(c)** $f(x) = \tan x$

FF11 Further Functions · original Q11 · source p.12

11

No calculator **Haese AA HL Ex 6A · Q11**

[Maximum mark: 3]

For what values of k is $f(x) = \cos(x - k)$:

(a) even **(b)** odd **(c)** neither?

FF9 Further Functions · original Q9 · source p.12

9 *Haese AA HL Ex 6A · Q9*

(a) $f(-x) = f(x) \Rightarrow ax^2 - bx + c = ax^2 + bx + c \Rightarrow 2bx = 0$ for all $x \Rightarrow \boxed{b = 0}$

(b) $g(-x) = -g(x)$: $-ax^3 + bx^2 - cx + d = -ax^3 - bx^2 - cx - d \Rightarrow 2bx^2 + 2d = 0$ for all x: $\boxed{b = 0 \text{ and } d = 0}$

(c) $h(-x) = h(x) \Rightarrow -2bx^3 - 2dx = 0$ for all x: $\boxed{b = 0 \text{ and } d = 0}$ (only even powers remain)

FF10 Further Functions · original Q10 · source p.12

10 *Haese AA HL Ex 6A · Q10*

(a) $\cos(-x) = \cos x$: **even** — graph symmetric about the y-axis

(b) $\sin(-x) = -\sin x$: **odd** — rotational symmetry (180°) about the origin

(c) $\tan(-x) = \dfrac{\sin(-x)}{\cos(-x)} = -\tan x$: **odd** — rotational symmetry about the origin

FF11 Further Functions · original Q11 · source p.12

11 *Haese AA HL Ex 6A · Q11*

(a) $\cos(x - k)$ is $\cos x$ translated; symmetric about the y-axis when $\boxed{k = n\pi,\ n \in \mathbb{Z}}$ ($f = \pm\cos x$)

(b) $f(x) = \pm\sin x$ when $\boxed{k = \dfrac{\pi}{2} + n\pi,\ n \in \mathbb{Z}}$ (e.g. $\cos(x - \dfrac{\pi}{2}) = \sin x$)

(c) **neither** for all other values: $k \neq \dfrac{n\pi}{2},\ n \in \mathbb{Z}$

FF12 Further Functions · original Q12 · source p.14

12

No calculator **Haese AA HL Ex 6A · Q12**

[Maximum mark: 6]

Prove that:

(a) the sum of two even functions is an even function

(b) the difference between two odd functions is an odd function

(c) the product of two odd functions is an even function.

FF13 Further Functions · original Q13 · source p.14

13

No calculator **Haese AA HL Ex 6A · Q13**

[Maximum mark: 2]

Suppose $f(x)$ is an even function and $g(x)$ is an odd function. Determine whether the function $(f \circ g)(x)$ is even, odd, or neither.

FF14 Further Functions · original Q14 · source p.14

14

No calculator **Haese AA HL Ex 6A · Q14**

[Maximum mark: 3]

$f(x)$ is an odd, invertible function. Determine whether the function $f^{-1}(x)$ is even, odd, or neither.

FF15 Further Functions · original Q15 · source p.16

15

No calculator **Haese Review 6A · Q1**

[Maximum mark: 3]

Determine whether each function is even, odd, or neither:

(a) $f(x) = -\dfrac{4}{x}$ **(b)** $f(x) = \dfrac{2x-3}{x+1}$ **(c)** $f(x) = \sqrt{x^2-5}$

FF12 Further Functions · original Q12 · source p.14

12 *Haese AA HL Ex 6A · Q12*

(a) $h = f+g$, f, g even: $h(-x) = f(-x)+g(-x) = f(x)+g(x) = h(x)$, so h is even.

(b) $h = f-g$, f, g odd: $h(-x) = -f(x)-(-g(x)) = -(f(x)-g(x)) = -h(x)$, so h is odd.

(c) $h = fg$, f, g odd: $h(-x) = (-f(x))(-g(x)) = f(x)g(x) = h(x)$, so h is even.

FF13 Further Functions · original Q13 · source p.14

13 *Haese AA HL Ex 6A · Q13*

$(f \circ g)(-x) = f(g(-x)) = f(-g(x))$ $\{g \text{ odd}\}$ **M1**

$= f(g(x))$ $\{f \text{ even}\}$ $= (f \circ g)(x)$: $\boxed{\text{even}}$ **A1**

FF14 Further Functions · original Q14 · source p.14

14 *Haese AA HL Ex 6A · Q14*

Let $y = f^{-1}(-x)$, so $f(y) = -x$. **M1**

f odd: $f(-y) = -f(y) = x$, so $-y = f^{-1}(x)$. **M1**

$f^{-1}(-x) = -f^{-1}(x)$: $\boxed{f^{-1} \text{ is odd}}$ (reflecting a graph with rotational symmetry in $y = x$ keeps the symmetry) **A1**

FF15 Further Functions · original Q15 · source p.16

15 *Haese Review 6A · Q1*

(a) $f(-x) = \dfrac{4}{x} = -f(x)$: **odd**

(b) $f(-x) = \dfrac{-2x-3}{1-x}$ — not $\pm f(x)$ (domain $x \neq -1$ not symmetric): **neither**

(c) $f(-x) = \sqrt{x^2-5} = f(x)$, domain $|x| \geq \sqrt{5}$ symmetric: **even**

FF16 Further Functions · original Q16 · source p.16

16

No calculator **Haese Review 6A · Q2**

[Maximum mark: 3]

For what values of k is $f(x) = \sin(x-k)$:

(a) even **(b)** odd **(c)** neither?

FF17 Further Functions · original Q17 · source p.16

17

No calculator **Haese Review 6A · Q3**

[Maximum mark: 2]

Given that $f(x)$ is an odd function, determine whether $(f \circ f)(x)$ is even, odd, or neither.

FF18 Further Functions · original Q18 · source p.16

18

No calculator **Haese Review 6A · Q4**

[Maximum mark: 3]

Suppose $f(x) = (x^2 - 3x)(x+b)$, $b \in \mathbb{R}$ is an odd function. Find the value of b.

FF19 Further Functions · original Q19 · source p.18

19

No calculator **Haese Review 6A · Q7**

[Maximum mark: 6]

Suppose $f(x)$ is an odd function. Determine whether the following are even, odd, or neither:

(a) $f(2x)$ **(b)** $f(x^2)$ **(c)** $f(x)+2$

(d) $[f(x)]^2$ **(e)** $|f(x)|$ **(f)** $\dfrac{1}{f(x)}$

FF16 Further Functions · original Q16 · source p.16

16 *Haese Review 6A · Q2*

(a) $f(x) = \pm\cos x$ when $\boxed{k = \frac{\pi}{2} + n\pi,\ n \in \mathbb{Z}}$

(b) $f(x) = \pm\sin x$ when $\boxed{k = n\pi,\ n \in \mathbb{Z}}$

(c) **neither** when $k \neq \frac{n\pi}{2},\ n \in \mathbb{Z}$

FF17 Further Functions · original Q17 · source p.16

17 *Haese Review 6A · Q3*

$(f \circ f)(-x) = f(f(-x)) = f(-f(x)) = -f(f(x))$ **M1**

$= -(f \circ f)(x)$: $\boxed{\text{odd}}$ **A1**

FF18 Further Functions · original Q18 · source p.16

18 *Haese Review 6A · Q4*

$f(x) = x^3 + (b-3)x^2 - 3bx$ **M1**

odd $\Rightarrow$ no even power: $b - 3 = 0$ **M1**

$\boxed{b = 3}$ (then $f(x) = x^3 - 9x$) **A1**

FF19 Further Functions · original Q19 · source p.18

19 *Haese Review 6A · Q7*

(a) $f(-2x) = -f(2x)$: **odd**

(b) $f((-x)^2) = f(x^2)$: **even**

(c) $f(-x) + 2 = -f(x) + 2$: **neither** (in general)

(d) $[f(-x)]^2 = [-f(x)]^2 = [f(x)]^2$: **even**

(e) $|f(-x)| = |-f(x)| = |f(x)|$: **even**

(f) $\dfrac{1}{f(-x)} = -\dfrac{1}{f(x)}$: **odd**

For part (f), $\frac{1}{f(x)}$ is odd on its natural symmetric domain $D = \{x \in \text{Dom}(f) : f(x) \neq 0\}$; exclude every zero of f.

FF20 Further Functions · original Q20 · source p.18

20

No calculator **Haese Review 6B · Q1**

[Maximum mark: 2]

Suppose $f(x)$ is an even function, and $(-2, 6)$ and $(1, -4)$ lie on the graph of $y = f(x)$. Write down two other points which lie on the graph of $y = f(x)$.

FF21 Further Functions · original Q21 · source p.18

21

No calculator **Haese Review 6B · Q2**

[Maximum mark: 4]

(a) Prove that the product of two even functions is an even function. [2]

(b) Give an example of two functions, both of which are neither odd nor even, whose product is an even function. [2]

FF22 Further Functions · original Q22 · source p.18

22

No calculator **Haese Review 6B · Q3**

[Maximum mark: 4]

$f(x) = \cos(2(x-a)) + b$ is an odd function. Find the possible values of a and b.

FF23 Further Functions · original Q23 · source p.20

23

No calculator **Haese Review 6B · Q5**

[Maximum mark: 2]

For any function $f(x)$, prove that $f(|x|)$ is an even function.

FF20 Further Functions · original Q20 · source p.18

20 *Haese Review 6B · Q1*

Reflect in the y-axis: $\boxed{(2, 6)}$ and $\boxed{(-1, -4)}$ **A1 A1**

FF21 Further Functions · original Q21 · source p.18

21 *Haese Review 6B · Q2*

(a) $h = fg$, f, g even: $h(-x) = f(-x)g(-x) = f(x)g(x) = h(x)$, so h is even. **M1 A1**

(b) e.g. $f(x) = x + 1$, $g(x) = x - 1$ (each neither): $f(x)g(x) = x^2 - 1$, which is even.

A2

FF22 Further Functions · original Q22 · source p.18

22 *Haese Review 6B · Q3*

Odd $\Rightarrow f(-x) = -f(x)$: $\cos(2x + 2a) + b = -\cos(2x - 2a) - b$ **M1**

$\cos(2x + 2a) + \cos(2x - 2a) = -2b \Rightarrow 2\cos 2x \cos 2a = -2b$ for all x **M1**

so $\cos 2a = 0$ and $b = 0$: $\boxed{a = \frac{\pi}{4} + \frac{n\pi}{2},\ n \in \mathbb{Z}; \quad b = 0}$ **A1 A1**

FF23 Further Functions · original Q23 · source p.20

23 *Haese Review 6B · Q5*

Let $h(x) = f(|x|)$. $h(-x) = f(|-x|) = f(|x|) = h(x)$ (since $|-x| = |x|$),

so h is even. **M1 A1**

FF24 Further Functions · original Q24 · source p.20

24

No calculator **Haese Review 6B · Q16**

[Maximum mark: 5]

Consider a function $f(x)$ with domain $x \in \mathbb{R}$.

(a) Determine whether each of the following is even, odd, or neither: (i) $\dfrac{f(x)+f(-x)}{2}$ (ii) $\dfrac{f(x)-f(-x)}{2}$ [3]

(b) Prove that $f(x)$ can be written as the sum of an even function and an odd function. [2]

FF25 Further Functions · original Q25 · source p.20

25

No calculator **Haese AA HL Ex 6B · Q5**

[Maximum mark: 2]

Show that if $f(x)$ is odd, then $[f(x)]^2$ is even.

FF26 Further Functions · original Q26 · source p.20

26

No calculator **Haese AA HL Ex 6C.1 · Q6**

[Maximum mark: 2]

Show that if $f(x)$ is an odd function, then $|f(x)|$ is an even function.

FF24 Further Functions · original Q24 · source p.20

24 *Haese Review 6B · Q16*

(a) (i) $E(x) = \dfrac{f(x)+f(-x)}{2}$: $E(-x) = \dfrac{f(-x)+f(x)}{2} = E(x)$: **even**

(ii) $O(x) = \dfrac{f(x)-f(-x)}{2}$: $O(-x) = \dfrac{f(-x)-f(x)}{2} = -O(x)$: **odd**

(b) $E(x)+O(x) = \dfrac{2f(x)}{2} = f(x)$, so f = (even) + (odd). e.g. $e^x = \cosh x + \sinh x$.

FF25 Further Functions · original Q25 · source p.20

25 *Haese AA HL Ex 6B · Q5*

$[f(-x)]^2 = [-f(x)]^2 = [f(x)]^2$, so $[f(x)]^2$ is even. **M1 A1**

FF26 Further Functions · original Q26 · source p.20

26 *Haese AA HL Ex 6C.1 · Q6*

$|f(-x)| = |-f(x)| = |f(x)|$, so $|f(x)|$ is even. **M1 A1**

FF40 Further Functions · original Q40 · source p.38

40

No calculator **Haese AA HL Ex 6C.1 · Q11**

[Maximum mark: 5]

Use the definition $|x| = \sqrt{x^2}$ to prove that:

(a) $|-x| = |x|$ for all x

(b) $|x|^2 = x^2$ for all x

(c) $|xy| = |x||y|$ for all x, y

(d) $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$, $y \neq 0$

(e) $|x-y| = |y-x|$ for all x, y

FF41 Further Functions · original Q41 · source p.40

41

No calculator **Haese AA HL Ex 6C.1 · Q12**

[Maximum mark: 8]

Use $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ to write the following functions without modulus signs. Hence graph each function.

(a) $y = |x-2|$

(b) $y = |x+1|$

(c) $y = -|x|$

(d) $y = |x| + x$

(e) $y = \frac{|x|}{x}$

(f) $y = x - 2|x|$

(g) $y = |x| + |x-2|$

(h) $y = |x| - |x-1|$

FF40 Further Functions · original Q40 · source p.38

40 *Haese AA HL Ex 6C.1 · Q11*

(a) $|-x| = \sqrt{(-x)^2} = \sqrt{x^2} = |x|$

(b) $|x|^2 = \left(\sqrt{x^2}\right)^2 = x^2$

(c) $|xy| = \sqrt{x^2y^2} = \sqrt{x^2}\sqrt{y^2} = |x||y|$

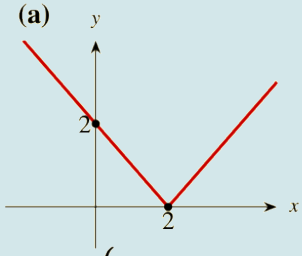
(d) $\left|\frac{x}{y}\right| = \sqrt{\frac{x^2}{y^2}} = \frac{\sqrt{x^2}}{\sqrt{y^2}} = \frac{|x|}{|y|}$

(e) $|x-y| = |-(y-x)| = |y-x|$ (by (a))

FF41 Further Functions · original Q41 · source p.40

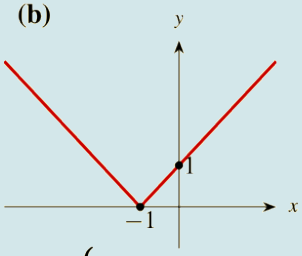
41 *Haese AA HL Ex 6C.1 · Q12*

(a)



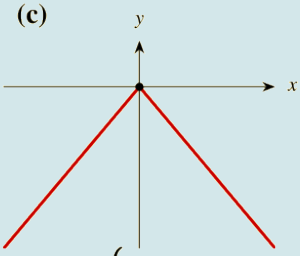
$$y = \begin{cases} x-2, & x \geq 2 \\ 2-x, & x < 2 \end{cases}$$

(b)



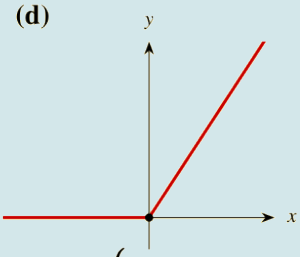
$$y = \begin{cases} x+1, & x \geq -1 \\ -x-1, & x < -1 \end{cases}$$

(c)



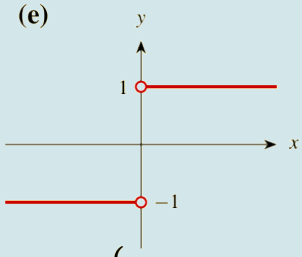
$$y = \begin{cases} -x, & x \geq 0 \\ x, & x < 0 \end{cases}$$

(d)



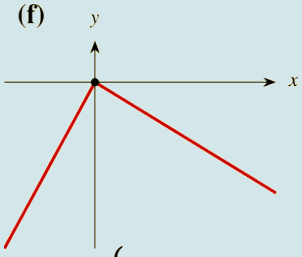
$$y = \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

(e)



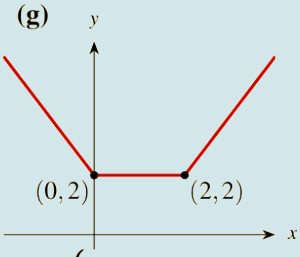
$$y = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

(f)



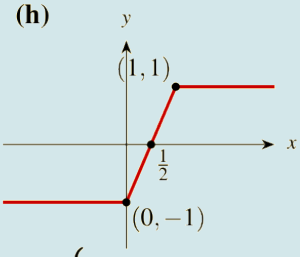
$$y = \begin{cases} -x, & x \geq 0 \\ 3x, & x < 0 \end{cases}$$

(g)



$$y = \begin{cases} 2-2x, & x < 0 \\ 2, & 0 \leq x < 2 \\ 2x-2, & x \geq 2 \end{cases}$$

(h)



$$y = \begin{cases} -1, & x < 0 \\ 2x-1, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

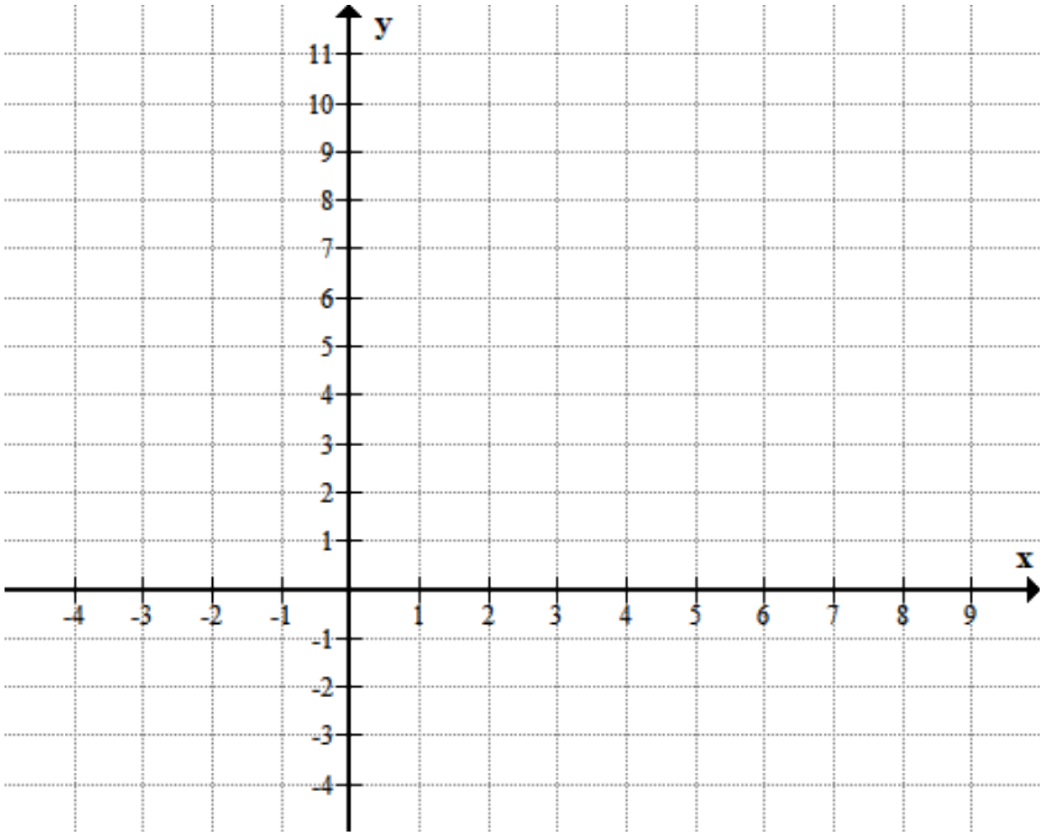
NK8 Christos Nikolaidis · Math-AA 2.15 · original Q8 · source p.7

8*. [Maximum mark: 8] ***[without GDC]***

(a) Draw the graph of $f(x) = |x-5| - |x| + 3$ [4]

(b) **Hence**, solve the inequalities

(i) $f(x) > 0$ (ii) $f(x) > 4$ (iii) $f(x) > -4$ (iv) $f(x) > 10$ [4]



NK8 Christos Nikolaidis · Math-AA 2.15 · original Q8 · source p.7

8. (a) The zeros of the moduli are $x = 0$ and $x = 5$. We find

$f(0) = 5 - 0 + 3 = 8$,

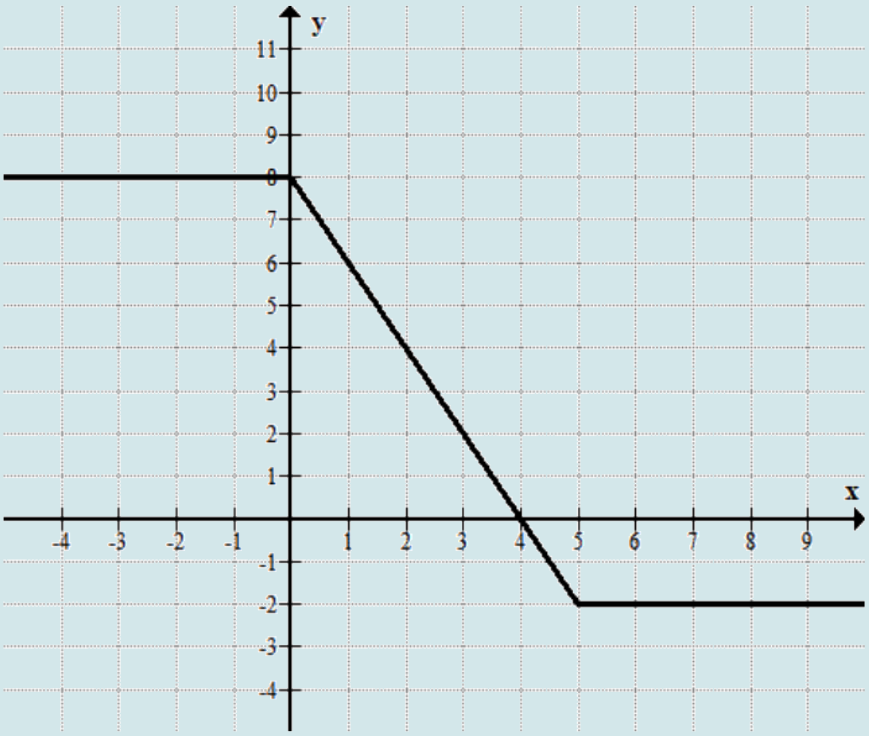
$f(5) = 0 - 5 + 3 = -2$

We also find one value before 0 and one value after 5.

$f(-1) = 6 - 1 + 3 = 8$,

$f(6) = 1 - 6 + 3 = -2$

We plot those points on the diagram and draw the graph.



(b) According to the graph

$f(x) > 0 \Leftrightarrow x < 4$

$f(x) > 4 \Leftrightarrow x < 2$

$f(x) > -4 \Leftrightarrow x \in \mathbb{R}$

$f(x) > 10$ no solution

FF31 Further Functions · original Q31 · source p.30

31

No calculator **Haese AA HL Ex 6C.1 · Q1**

[Maximum mark: 4]

Draw $y = f(x) = 2x - 4$, and on the same set of axes draw the graph of:

(a) $y = |f(x)|$ **(b)** $y = f(|x|)$

FF32 Further Functions · original Q32 · source p.30

32

No calculator **Haese AA HL Ex 6C.1 · Q2**

[Maximum mark: 4]

Draw $y = f(x) = x(x+2)$, and on the same set of axes draw the graph of:

(a) $y = |f(x)|$ **(b)** $y = f(|x|)$

FF33 Further Functions · original Q33 · source p.32

33

No calculator **Haese AA HL Ex 6C.1 · Q3**

[Maximum mark: 4]

Draw $y = f(x) = -x^2 + 6x - 8$, and on the same set of axes draw the graph of:

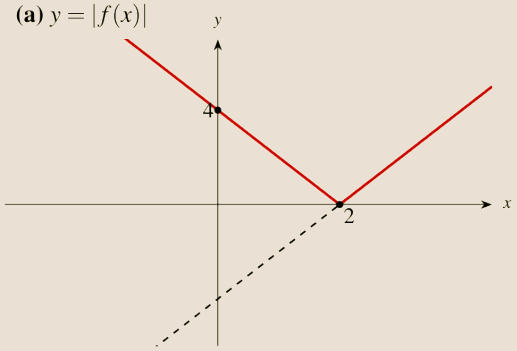
(a) $y = |f(x)|$ **(b)** $y = f(|x|)$

FF31 Further Functions · original Q31 · source p.30

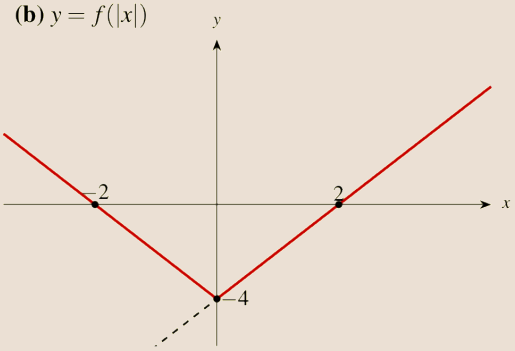
31 *Haese AA HL Ex 6C.1 · Q1*

(dashed: $y = f(x)$; red: new graph)

(a) $y = |f(x)|$



(b) $y = f(|x|)$

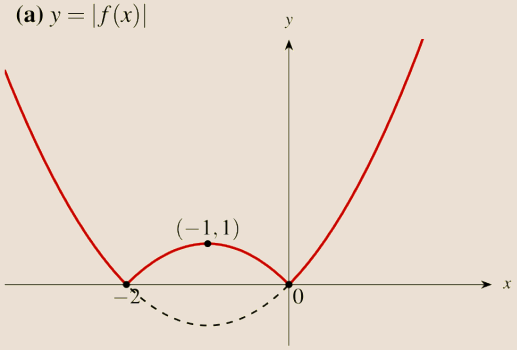


FF32 Further Functions · original Q32 · source p.30

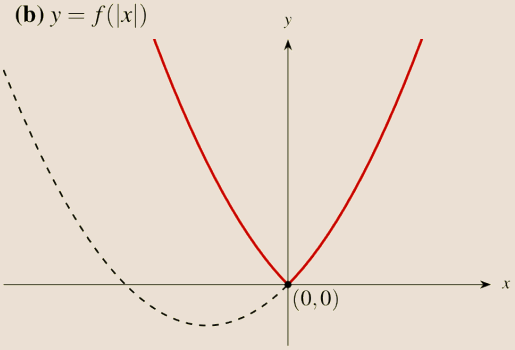
32 *Haese AA HL Ex 6C.1 · Q2*

(dashed: $y = f(x)$; red: new graph)

(a) $y = |f(x)|$



(b) $y = f(|x|)$

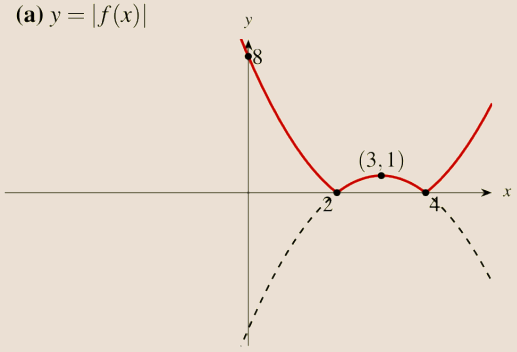


FF33 Further Functions · original Q33 · source p.32

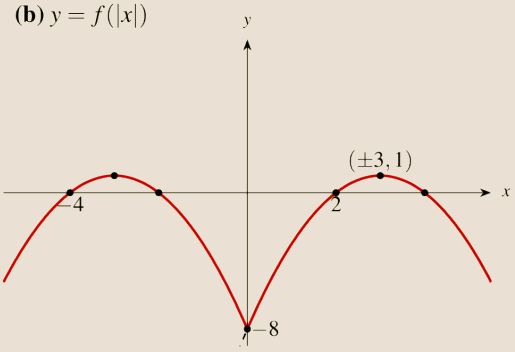
33 *Haese AA HL Ex 6C.1 · Q3*

(dashed: $y = f(x)$; red: new graph)

(a) $y = |f(x)|$



(b) $y = f(|x|)$



FF34 Further Functions · original Q34 · source p.32

34

No calculator

Haese AA HL Ex 6C.1 · Q4

[Maximum mark: 4]

Draw $y = f(x) = 3^x - 3$, and on the same set of axes draw the graph of:

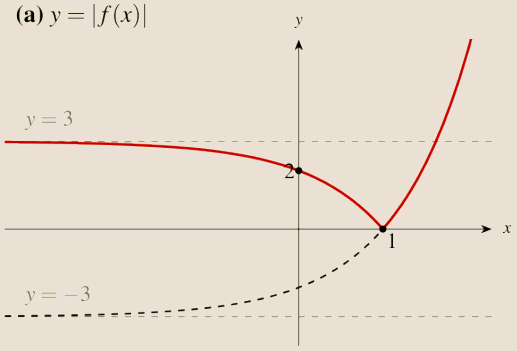
(a) $y = |f(x)|$ **(b)** $y = f(|x|)$

FF34 Further Functions · original Q34 · source p.32

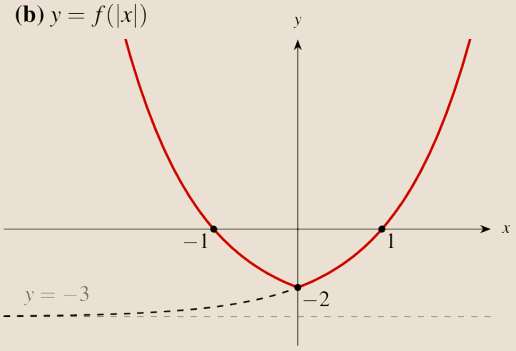
34 *Haese AA HL Ex 6C.1 · Q4*

(dashed: $y = f(x)$; red: new graph)

(a) $y = |f(x)|$



(b) $y = f(|x|)$



FF35 Further Functions · original Q35 · source p.34

35

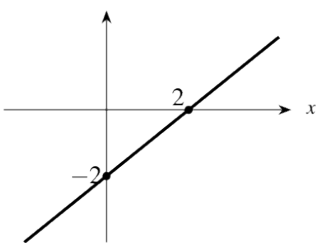
No calculator

Haese AA HL Ex 6C.1 · Q5

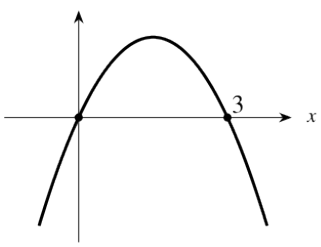
[Maximum mark: 8]

(a) Copy the following graphs for $y = f(x)$ and on the same set of axes draw the graph of $y = |f(x)|$:

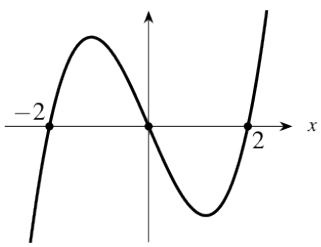
(i)



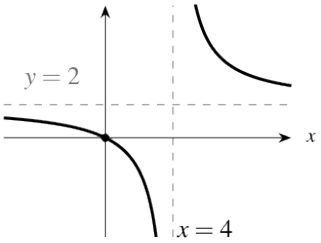
(ii)



(iii)



(iv)



(b) Repeat (a), but this time draw the graphs of $y = f(x)$ and $y = f(|x|)$ on the same set of axes.

FF36 Further Functions · original Q36 · source p.36

36

No calculator

Haese AA HL Ex 6C.1 · Q7

[Maximum mark: 6]

Let $f(x) = \sqrt{4 - x}$.

(a) Draw the graph of $y = f(x)$ and state its domain and range. [3]

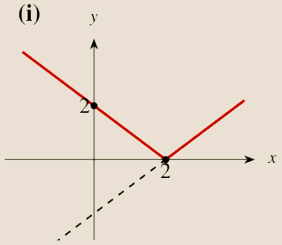
(b) Draw the graph of $y = f(|x|)$ and state its domain and range. [3]

FF35 Further Functions · original Q35 · source p.34

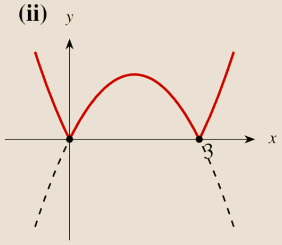
35 *Haese AA HL Ex 6C.1 · Q5*

(a) $y = |f(x)|$ (dashed: $y = f(x)$; red: new graph)

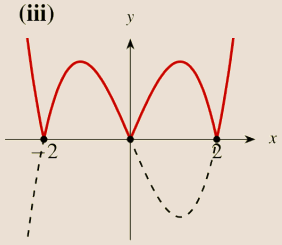
(i)



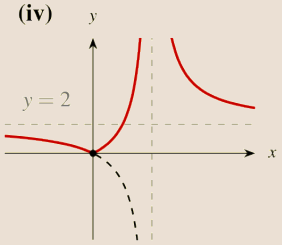
(ii)



(iii)

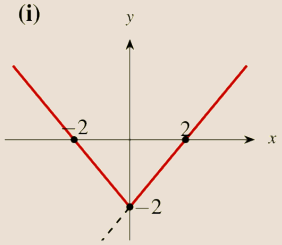


(iv)

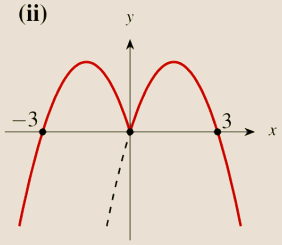


(b) $y = f(|x|)$: keep the part for $x \geq 0$ and reflect it in the y-axis.

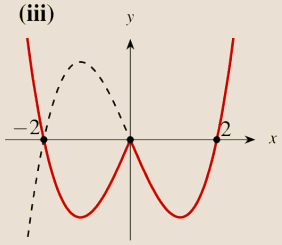
(i)



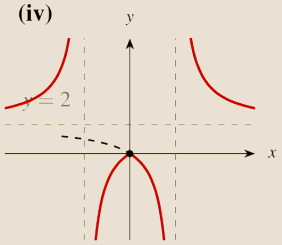
(ii)



(iii)



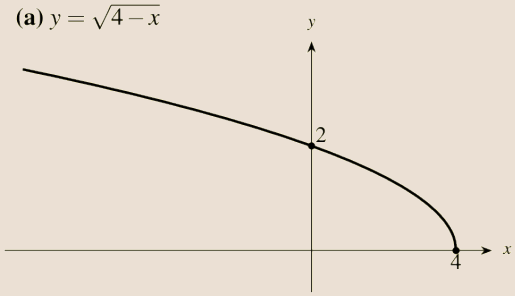
(iv)



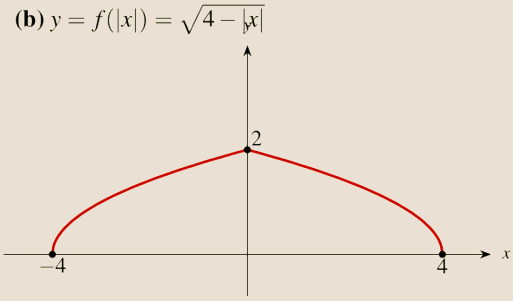
FF36 Further Functions · original Q36 · source p.36

36 *Haese AA HL Ex 6C.1 · Q7*

(a) $y = \sqrt{4 - x}$



(b) $y = f(|x|) = \sqrt{4 - |x|}$



(a) Domain $\boxed{x \leq 4}$, range $\boxed{y \geq 0}$ **A1 A1**

(b) Domain $\boxed{-4 \leq x \leq 4}$, range $\boxed{0 \leq y \leq 2}$ **A1 A1**

FF37 Further Functions · original Q37 · source p.36

37

No calculator **Haese AA HL Ex 6C.1 · Q8**

[Maximum mark: 4]

Suppose $f(x)$ has domain $-2 \leq x \leq 6$ and range $-7 \leq y \leq 5$.

(a) Find: (i) the domain of $f(|x|)$ (ii) the range of $|f(x)|$. [2]

(b) Can we determine the range of $f(|x|)$? Explain your answer. [2]

FF38 Further Functions · original Q38 · source p.36

38

No calculator **Haese AA HL Ex 6C.1 · Q9**

[Maximum mark: 4]

Suppose $f(x)$ has x-intercepts -3 and 4, and y-intercept -2. Find the axes intercepts of:

(a) $|f(x)|$ **(b)** $f(|x|)$

FF39 Further Functions · original Q39 · source p.38

39

No calculator **Haese AA HL Ex 6C.1 · Q10**

[Maximum mark: 4]

Suppose $f(x) = \log_2(x+4)$. Sketch the graphs of $y = f(x)$ and $y = |f(|x|)|$ on the same set of axes.

FF37 Further Functions · original Q37 · source p.36

37 *Haese AA HL Ex 6C.1 · Q8*

(a)(i) need $|x| \leq 6$: $\boxed{-6 \leq x \leq 6}$ **A1** (ii) $\boxed{0 \leq y \leq 7}$ **A1**

(b) **No.** $f(|x|)$ only uses the values of f for $0 \leq x \leq 6$; we do not know which part of the range $-7 \leq y \leq 5$ comes from $-2 \leq x < 0$. **R2**

FF38 Further Functions · original Q38 · source p.36

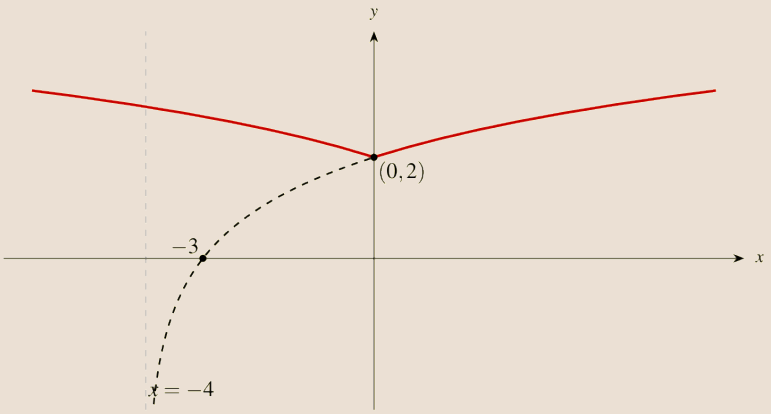
38 *Haese AA HL Ex 6C.1 · Q9*

(a) x-intercepts $\boxed{-3 \text{ and } 4}$, y-intercept $\boxed{2}$ **A1 A1**

(b) uses $x \geq 0$ only: x-intercepts $\boxed{\pm 4}$, y-intercept $\boxed{-2}$ ($x = -3$ is lost) **A1 A1**

FF39 Further Functions · original Q39 · source p.38

39 *Haese AA HL Ex 6C.1 · Q10*



$f(|x|) = \log_2(|x|+4) \geq \log_2 4 = 2 > 0$, so the outer modulus changes nothing: $y = |f(|x|)| = \log_2(|x|+4)$, an even function with minimum $(0, 2)$, no x-intercepts and no asymptote. **A2 A2**

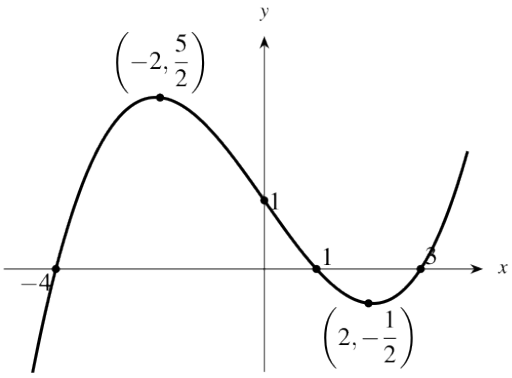
FF42 Further Functions · original Q42 · source p.42

42

No calculator **Haese Review 6A · Q5(a)**

[Maximum mark: 3]

The graph of $y = f(x)$ is given. On the same set of axes, graph the functions $y = f(x)$ and $y = f(|x|)$.



(part (b), $y = [f(x)]^2$, is in Section 4)

FF43 Further Functions · original Q43 · source p.42

43

No calculator **Haese Review 6A · Q6**

[Maximum mark: 2]

Suppose $f(x)$ has domain $0 \leq x < 5$ and range $-5 \leq y \leq 2$. Find:

(a) the domain of $f(|x|)$ **(b)** the range of $|f(x)|$

FF44 Further Functions · original Q44 · source p.42

44

No calculator **Haese Review 6A · Q9**

[Maximum mark: 4]

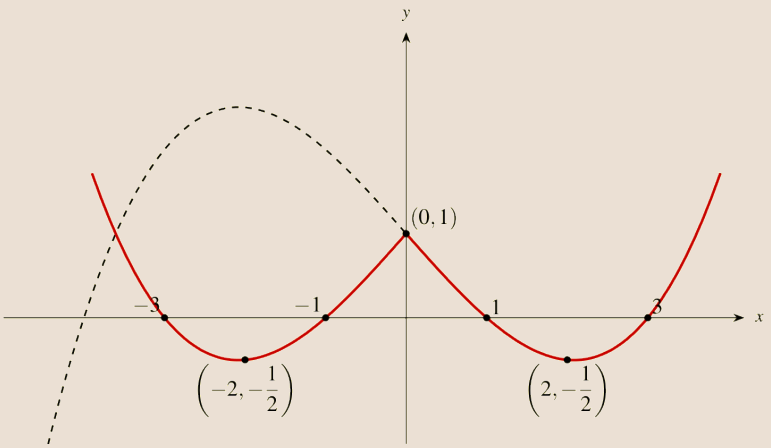
Draw the graph of:

(a) $y = |(x-1)(x-4)|$ **(b)** $y = 2x - |x-2|$

FF42 Further Functions · original Q42 · source p.42

42 *Haese Review 6A · Q5(a)*

(dashed: $y = f(x)$; red: new graph)



$f(|x|)$ keeps the right half ($x \geq 0$) and reflects it in the y-axis: x-intercepts $\pm 1, \pm 3$, minima $(\pm 2, -\frac{1}{2})$, sharp maximum $(0, 1)$. **A3**

FF43 Further Functions · original Q43 · source p.42

43 *Haese Review 6A · Q6*

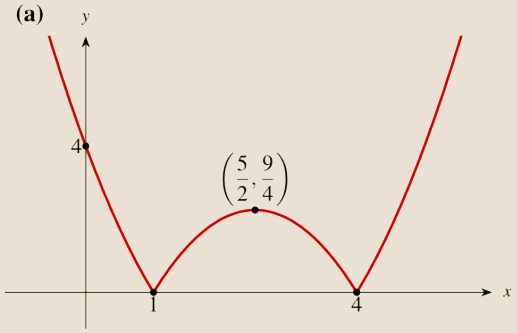
(a) $\boxed{-5 < x < 5}$ **A1**

(b) $\boxed{0 \leq y \leq 5}$ **A1**

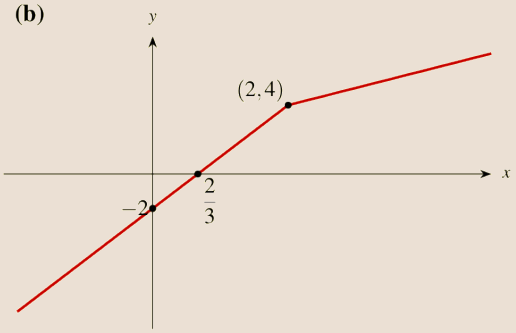
FF44 Further Functions · original Q44 · source p.42

44 *Haese Review 6A · Q9*

(a)



(b)



(b) $y = \begin{cases} x + 2, & x \geq 2 \\ 3x - 2, & x < 2 \end{cases}$ (two straight pieces meeting at $(2, 4)$) **A2**

FF45 Further Functions · original Q45 · source p.44

45

No calculator **Haese Review 6A · Q13**

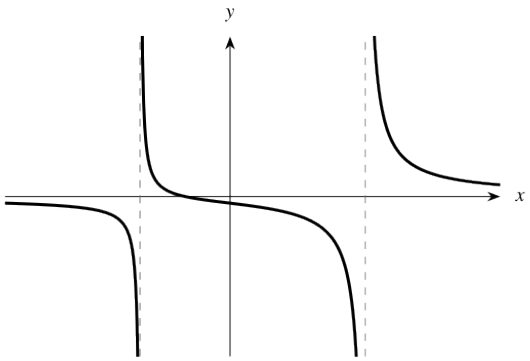
[Maximum mark: 6]

The graph of $f(x) = \dfrac{x+1}{x^2-x-6}$ is shown alongside.

(a) Find the asymptotes of the function. [2]

(b) Find the axes intercepts. [2]

(c) Sketch the graph of $y = |f(x)|$. [2]



FF46 Further Functions · original Q46 · source p.44

46

No calculator **Haese Review 6B · Q6**

[Maximum mark: 3]

The graph of $y = f(x)$ has 3 turning points and 2 x-intercepts. If possible, state how many turning points and x-intercepts each of the following graphs has:

(a) $y = |f(x)|$ **(b)** $y = f(|x|)$ **(c)** $y = [f(x)]^2$

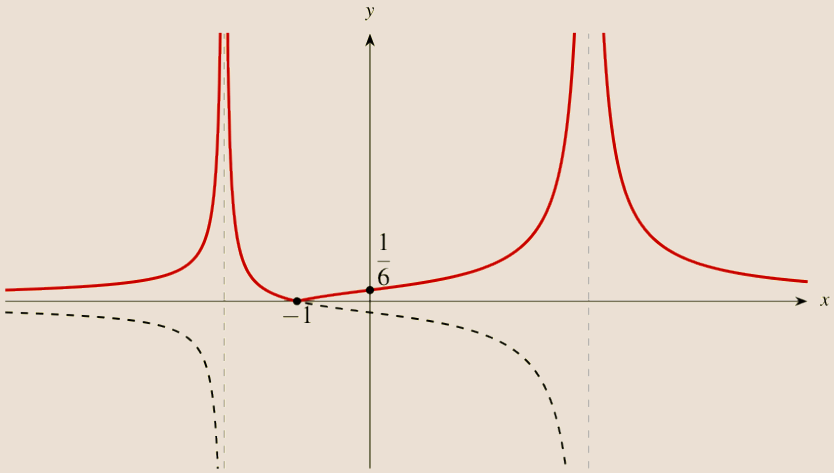
FF45 Further Functions · original Q45 · source p.44

45 *Haese Review 6A · Q13*

(a) $x^2 - x - 6 = (x-3)(x+2)$: V.A. $\boxed{x = -2,\ x = 3}$; H.A. $\boxed{y = 0}$ **A1 A1**

(b) x-intercept $\boxed{-1}$, y-intercept $\boxed{-\frac{1}{6}}$ **A1 A1**

(c) **A2**



FF46 Further Functions · original Q46 · source p.44

(a) For $y = |f(x)|$, the same two zero locations remain. Every crossing of the original x-axis becomes a cusp minimum. Original turning points remain, but an axis-touching turning point must not be counted again. Without knowing which zeros cross, the exact turning-point count cannot be fixed; in the ordinary isolated-turn case the possible counts are 3, 4 or 5. (b) For $y = f(|x|)$, only the part with $x \geq 0$ is retained and reflected. Neither the number of turning points nor the number of x-intercepts is determined by the stated whole-graph counts. (c) For $y = (f(x))^2$, the same two zero locations remain. Crossings become minima; touching zeros already counted as turning points are not added twice. The exact turning-point count is again undetermined without the graph's local shape. **Clarification:** the source answer's alternatives 3 and 5 omit the possibility of one crossing and one tangency.

FF47 Further Functions · original Q47 · source p.46

47

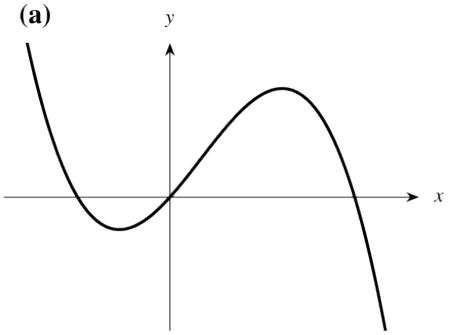
No calculator

Haese Review 6B · Q7

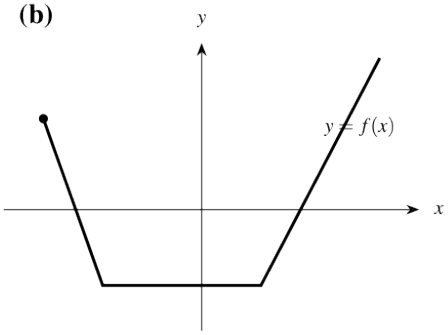
[Maximum mark: 8]

Draw the graph of $y = |f(x)|$ and $y = f(|x|)$ for:

(a)



(b)

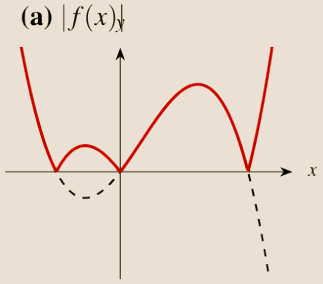


FF47 Further Functions · original Q47 · source p.46

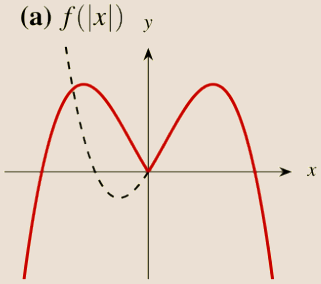
47 *Haese Review 6B · Q7*

(dashed: $y = f(x)$; red: new graph)

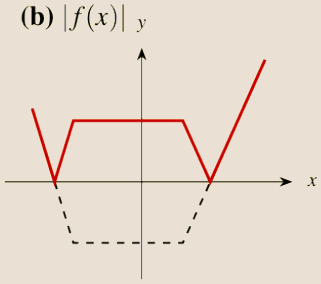
(a) $|f(x)|$



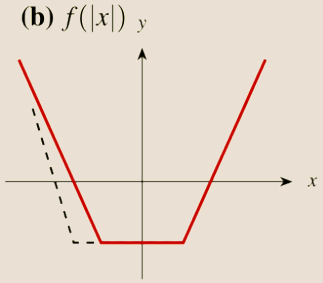
(a) $f(|x|)$



(b) $|f(x)|$



(b) $f(|x|)$



$|f(x)|$: parts below the x-axis are reflected upwards. $f(|x|)$: the part for $x \geq 0$ is kept and mirrored in the y-axis (the left-hand part of f is discarded). **A2 A2 A2 A2**

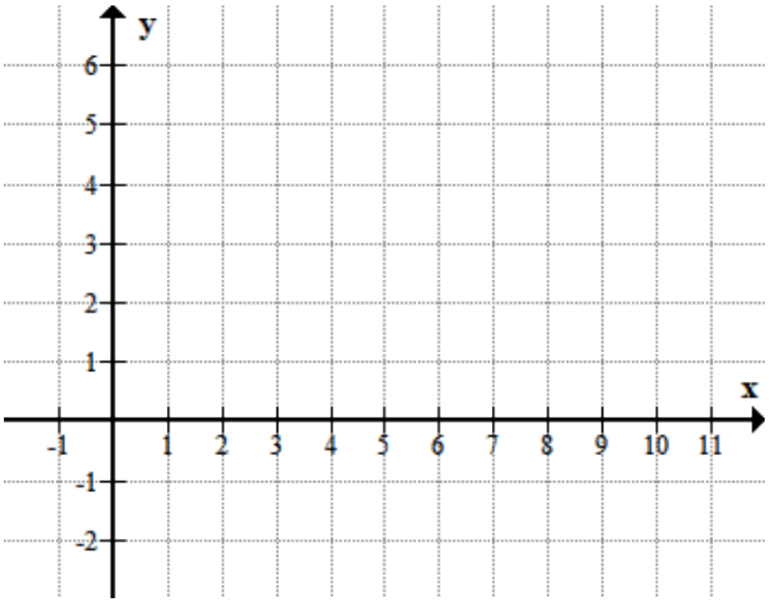
NK3 Christos Nikolaidis · Math-AA 2.15 · original Q3 · source p.3

3. [Maximum mark: 7] ***[with GDC]***

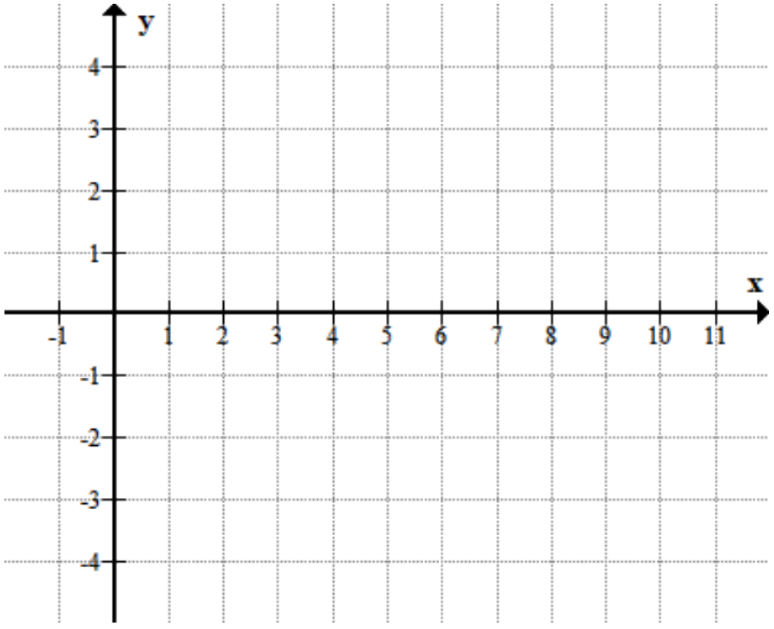
(a) (Using SolveN or the graph mode on your GDC), solve the equation $|x-5|=3$. [2]

(b) (Using the graph mode, solve the inequalities (i) $|x-5|<3$ (ii) $|x-5|>3$ [4]

METHOD A: Draw the graphs of $y_1=|x-5|$ and $y_2=3$



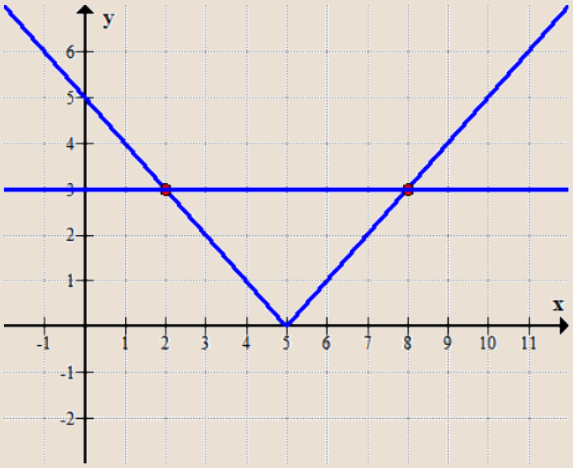
METHOD B: Draw the graph of $y=|x-5|-3$



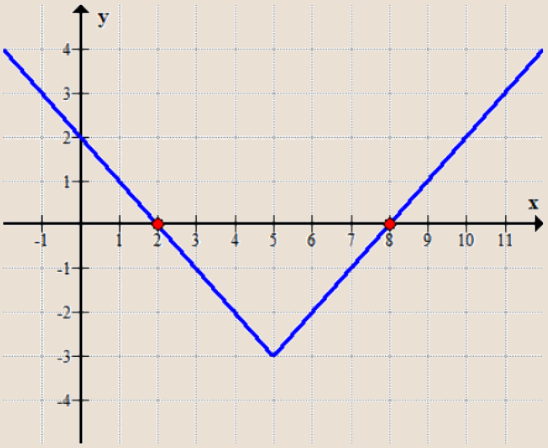
NK3 Christos Nikolaidis · Math-AA 2.15 · original Q3 · source p.3

3. SolveN gives directly the solutions $x = 8,\ x = 2$

The graphs are as follows:



$y_1 = |x-5|$ and $y_2 = 3$



$y = |x-5|-3$

1st diagram: we find the points of intersection and observe where one graph is above the other

2nd diagram: we find the roots and observe where the graph is above or below the x-axis.

(a) $x = 8,\ x = 2$

(b) $2 < x < 8$

(c) $x < 2$ or $x > 8$

NK16 Christos Nikolaidis · Math-AA 2.15 · original Q16 · source p.12

16. [Maximum mark: 14] ***[without GDC]***

Let $f(x) = |x-3| + |x+2|$

(a) Draw the graph of the function f in the diagram below. [4]

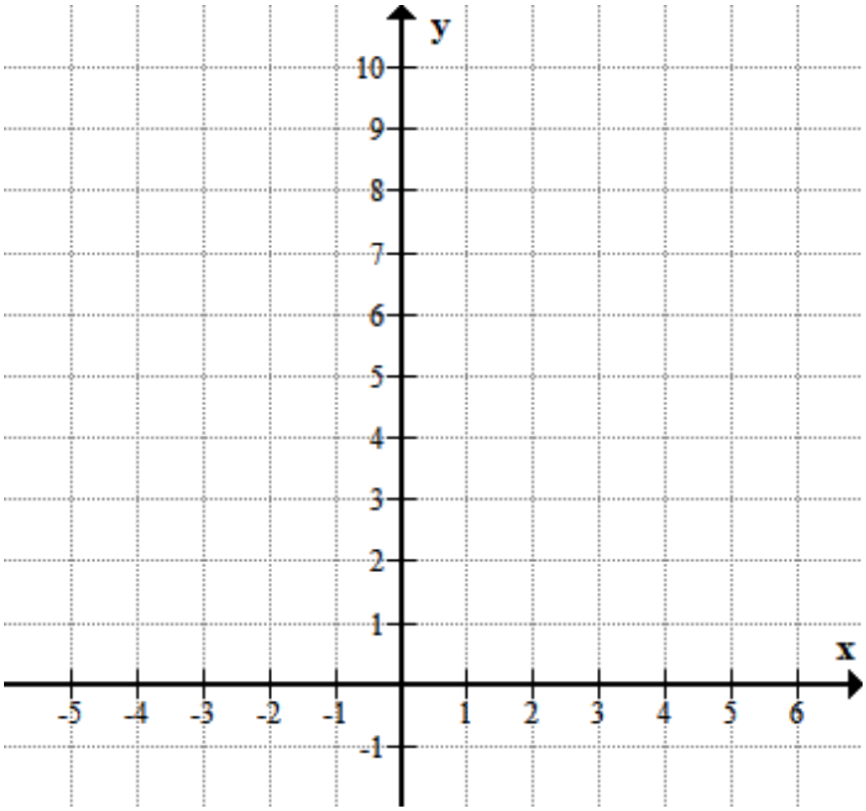
(b) Write down the range of f. [2]

(c) Express the function in the form $f(x) = \begin{cases} ax+b & x < -2 \\ c & -2 \le x < 3 \\ dx+e & x \ge 3 \end{cases}$ [4]

(d) Hence, or otherwise,

(i) solve the equation $f(x) = 9$;

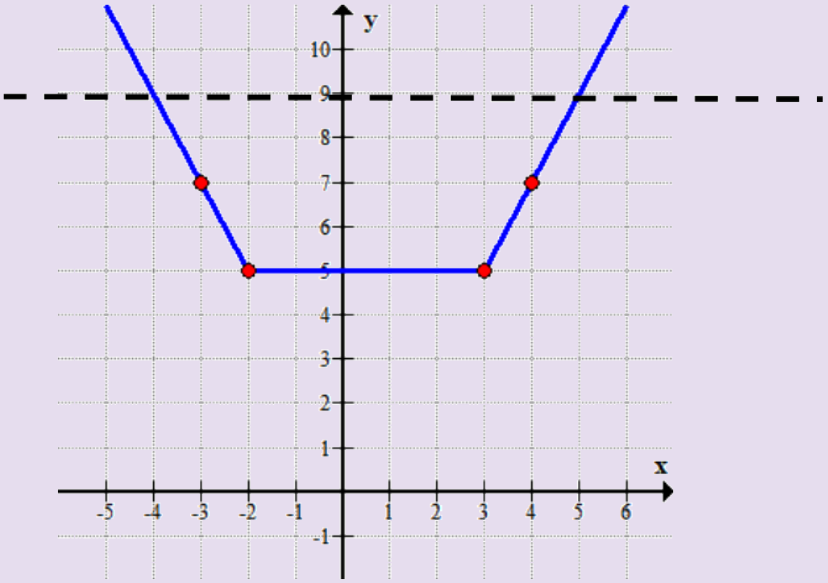
(ii) solve the inequality $f(x) \le 9$. [4]



NK16 Christos Nikolaidis · Math-AA 2.15 · original Q16 · source p.12

16. (a) We consider the zeros of abs. values: $-2,\ 3$, one point before one point after, say $-3, 4$

$f(-2)=5$, $f(3)=5$ $f(-3)=7$ $f(4)=7$



(b) $y \geq 5$

(c) For $x \geq 3$, $f(x)=(x-3)+(x+2)=2x-1$

For $-2 \leq x < 3$, $f(x)=-(x-3)+(x+2)=5$

For $x < -2$, $f(x)=-(x-3)-(x+2)=-2x+1$

$$f(x)=\begin{cases} -2x+1 & x<-2 \\ 5 & -2\leq x<3 \\ 2x-1 & x\geq 3 \end{cases}$$

(d) (i) **EITHER** directly from the graph: $x=-4$, $x=5$

OR solving $-2x+1=9 \Leftrightarrow x=-4$ and $2x-1=9 \Leftrightarrow x=5$

(ii) $-4 \leq x \leq 5$

NK17 Christos Nikolaidis · Math-AA 2.15 · original Q17 · source p.14

17. [Maximum mark: 14] ***[without GDC]***

Let $f(x) = |x-3| - |x+2|$

(a) Draw the graph of the function f in the diagram below. [4]

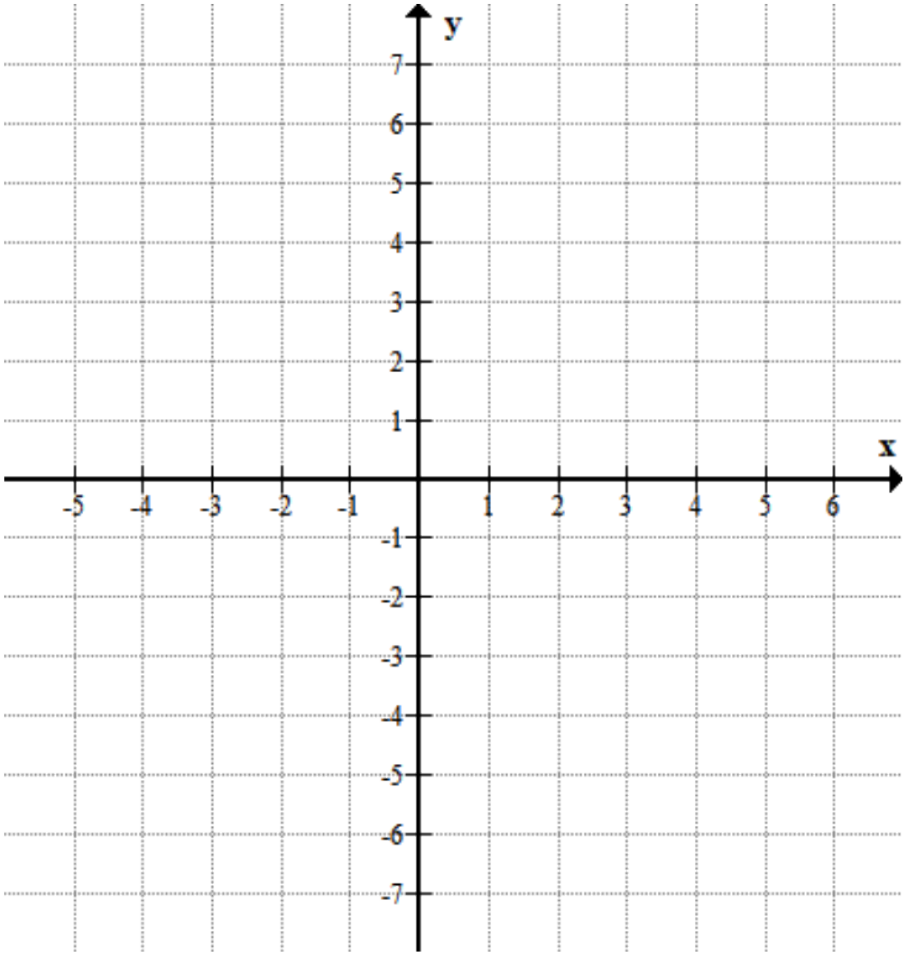
(b) Write down the range of f. [2]

(c) Express the function in the form $f(x) = \begin{cases} a & x < -2 \\ bx + c & -2 \le x < 3 \\ d & x \ge 3 \end{cases}$ [4]

(d) Hence, or otherwise,

(i) solve the equation $f(x) = 0$;

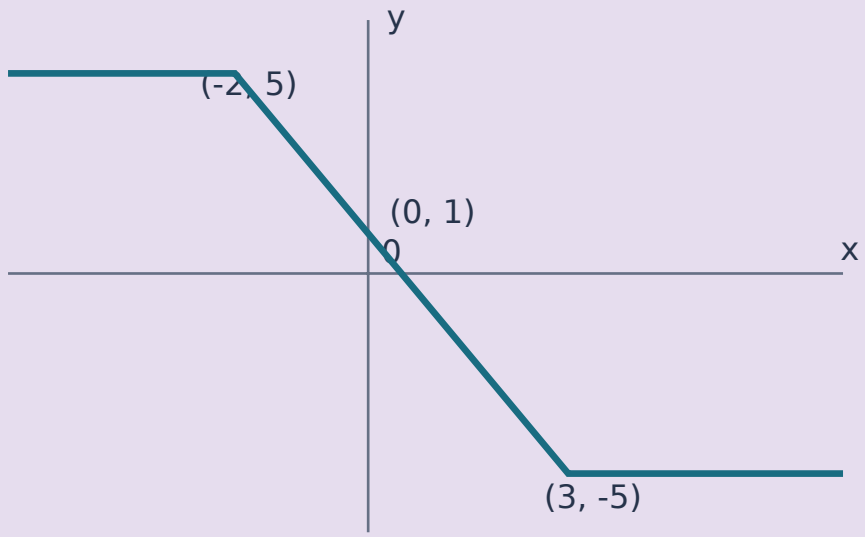
(ii) solve the inequality $f(x) > 1$. [4]



NK17 Christos Nikolaidis · Math-AA 2.15 · original Q17 · source p.14

(a) The change points are $x = -2$ and $x = 3$:

$$f(x) = \begin{cases} 5 & \text{if } x \leftarrow 2 \\ 1 - 2x & \text{if } -2 \leq x < 3. \\ -5 & \text{if } x \geq 3 \end{cases}$$



The horizontal branches meet the sloping segment at $(-2, 5)$ and $(3, -5)$. (b) Range: $-5 \leq y \leq 5$. (c) In the requested casewise form, $a = 5$, $b = -2$, $c = 1$, $d = -5$. (d)(i) $1 - 2x = 0$ gives $x = \frac{1}{2}$ (which lies in the middle interval). (ii) On the middle interval, $1 - 2x > 1$ gives $x < 0$. The left branch also satisfies the inequality; the right branch does not. Thus $\boldsymbol{x < 0}$. Check: $f(4) = |4 - 3| - |4 + 2| = 1 - 6 = -5$. The source key's printed value $+5$ at $x = 4$ is a typographical error.

FF27 Further Functions · original Q27 · source p.22

27

Calculator **RV · Q1**

[Maximum mark: 5]

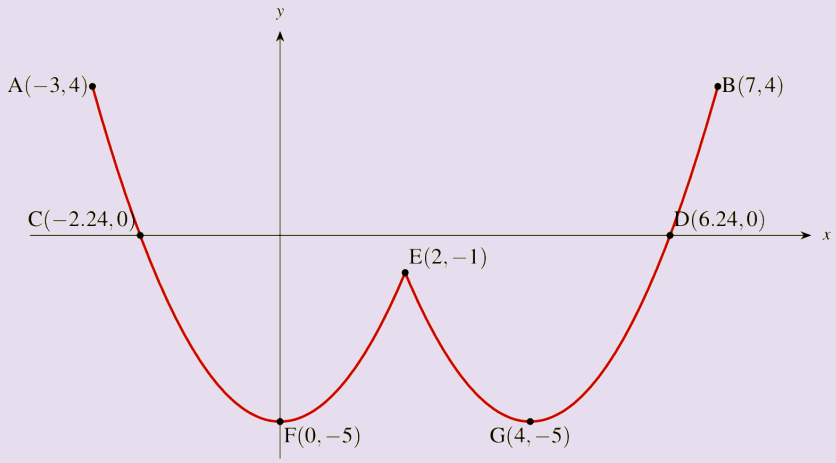
(a) Sketch the graph of $y = (x-2)^2 - 4|x-2| - 1$, for $-3 \leq x \leq 7$. [3]

(b) Hence, or otherwise, solve the equation $(x-2)^2 - 4|x-2| - 1 = 0$. [2]

FF27 Further Functions · original Q27 · source p.22

27 *RV · Q1*

(a) **A1 A1 A1**



Symmetric about $x = 2$ (it is a function of $|x-2|$); minima where $|x-2| = 2$, cusp at $x = 2$. Award: shape, endpoints, intercepts/turning points.

(b) Let $u = |x-2|$: $u^2 - 4u - 1 = 0 \Rightarrow u = 2+\sqrt{5}$ (reject $2-\sqrt{5} < 0$) **M1**

$x = 2 \pm (2+\sqrt{5})$: $\boxed{x = -\sqrt{5} \approx -2.24, \quad x = 4+\sqrt{5} \approx 6.24}$ **A1**

TI-Nspire CX: Question 27

`f1(x)=(x-2)^2-4abs(x-2)-1 | -3≤x≤7.` menu → **6: Analyze Graph** → **1: Zero** gives x-intercepts; **2: Minimum** / **3: Maximum** give turning points. Zeros: -2.24 and 6.24; minima $(0,-5)$, $(4,-5)$.

FF28 Further Functions · original Q28 · source p.24

28

Calculator RV · Q2

[Maximum mark: 7]

Let $f(x) = \dfrac{4 - ax}{5x - 2}$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$. [1]

The line $y = -1.4$ is a horizontal asymptote to the graph of $y = f(x)$.

(b) Find the value of a. [2]

(c) The line $y = x - c$, where $c \in \mathbb{R}^+$, intersects the graph of $y = |f(x)|$ at exactly one point. Find the possible values of c. [4]

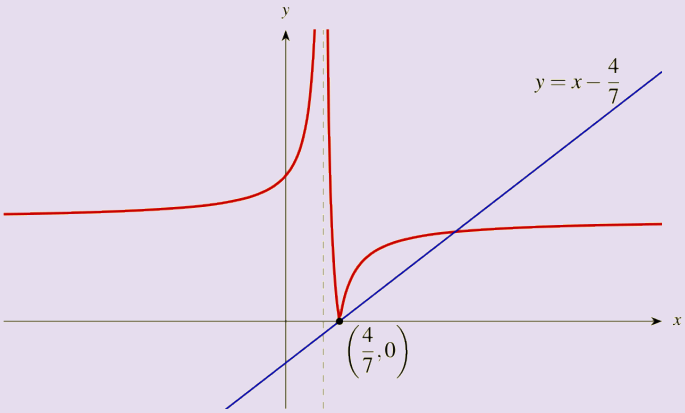
FF28 Further Functions · original Q28 · source p.24

28 *RV · Q2*

(a) $5x-2=0$: $\boxed{x=\frac{2}{5}}$ **A1**

(b) H.A. $y=-\frac{a}{5}=-1.4$ **M1** $\boxed{a=7}$ **A1**

(c) $|f(x)|=\left|\frac{4-7x}{5x-2}\right|$, zero at $x=\frac{4}{7}\approx 0.571$; H.A. $y=1.4$. **M1**



A line of gradient 1 through $\left(\frac{4}{7},0\right)$ meets the graph twice; any line of gradient 1 lower than this meets it once only **R1 M1**

$\boxed{c>\frac{4}{7}\ (\approx 0.571)}$ **A1**

TI-Nspire CX: Question 28

`f1(x)=abs((4-7x)/(5x-2))`, `f2(x)=x-c` with a slider for c (menu → **1: Actions** → **B: Insert Slider**). menu → **6: Analyze Graph** → **1: Zero** gives x-intercepts; **2: Minimum / 3: Maximum** give turning points. Zero of f1: $x=0.571$.

FF29 Further Functions · original Q29 · source p.26

29

No calculator **RV · Q5**

[Maximum mark: 5]

Consider the graphs of $y = -\frac{1}{2}|x|$ and $y = 2|x| - a$, where $a \in \mathbb{Z}^+$.

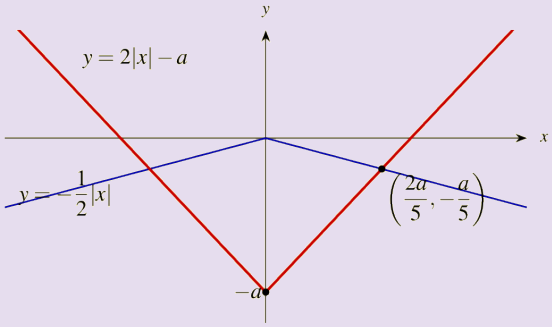
(a) Sketch the graphs on the same set of axes. [2]

(b) Given that the graphs enclose a region of area 40 square units, find the value of a. [3]

FF29 Further Functions · original Q29 · source p.26

29 *RV · Q5*

(a) **A1 A1**



(b) For $x > 0$: $-\frac{1}{2}x = 2x - a \Rightarrow x = \frac{2a}{5}$ **M1**

Region = 2 triangles, base a (on the y-axis), height $\frac{2a}{5}$: Area $= 2 \times \frac{1}{2} \times a \times \frac{2a}{5} = \frac{2a^2}{5}$ **M1**

$\frac{2a^2}{5} = 40 \Rightarrow a^2 = 100$, $\boxed{a = 10}$ $(a > 0)$ **A1**

FF30 Further Functions · original Q30 · source p.28

30

Calculator **RV · Q6**

[Maximum mark: 7]

Let $f(x) = \dfrac{9-12x}{cx-20}$, for $x \neq \dfrac{20}{c}$, where $c \neq 0$.

(a) The line $x = 5$ is a vertical asymptote to the graph of $y = f(x)$.

(i) Find the value of c.

(ii) Write down the equation of the horizontal asymptote to the graph of $y = f(x)$. [4]

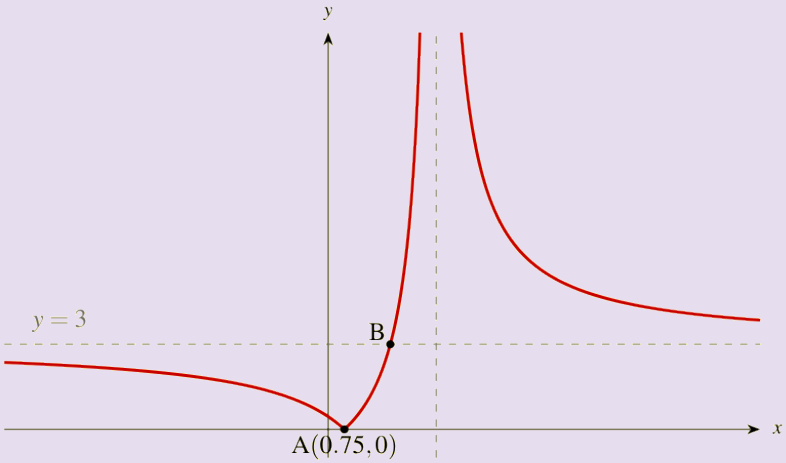
(b) The line $y = h$, where $h \in \mathbb{R}$, intersects the graph of $y = |f(x)|$ at exactly one point. Find the possible values of h. [3]

FF30 Further Functions · original Q30 · source p.28

30 *RV · Q6*

(a)(i) $5c - 20 = 0$ **M1** $\boxed{c = 4}$ **A1** (ii) $y = \dfrac{-12}{4}$: $\boxed{y = -3}$ **A1 A1**

(b)



$|f(x)|$ has H.A. $y = 3$. $y = h$ meets the graph once only at the minimum A ($h = 0$) or at B: $|f(x)| = 3$ has only one solution, $x = 2.875$ (since $f(x) = -3$ is impossible). **M1 R1**

$\boxed{h = 0 \text{ or } h = 3}$ **A1**

TI-Nspire CX: Question 30

`f1(x)=abs((9-12x)/(4x-20))`, `f2(x)=3`. Intersection $\rightarrow (2.875, 3)$; Zero $\rightarrow x = 0.75$.

NK1 Christos Nikolaidis · Math-AA 2.15 · original Q1 · source p.1

1. [Maximum mark: 15] ***[without GDC]***

(a) Solve the following equations and inequalities [using the definition of $|x|$] [9]

$\|x\| = 3$	
$\|x\| < 3$	
$\|x\| > 3$	
$\|x\| = -3$	
$\|x\| < -3$	
$\|x\| > -3$	
$\|x\| = 0$	
$\|x\| < 0$	
$\|x\| > 0$	

(b) Solve the following equation and inequalities ` [6]

$2\|x\| - 3 = 0$	
$2\|x\| - 3 < 0$	
$2\|x\| - 3 > 0$	

NK1 Christos Nikolaidis · Math-AA 2.15 · original Q1 · source p.1

Corrected and checked worked answer. (a) Use $|x| \geq 0$ and the definition of absolute value:

$\lvert x\rvert = 3$	$x = -3$ or $x = 3$
$\lvert x\rvert < 3$	$-3 < x < 3$
$\lvert x\rvert > 3$	$x \leftarrow 3$ or $x > 3$
$\lvert x\rvert = -3$	No solution
$\lvert x\rvert \leftarrow 3$	No solution
$\lvert x\rvert > -3$	All real x
$\lvert x\rvert = 0$	$x = 0$
$\lvert x\rvert < 0$	No solution
$\lvert x\rvert > 0$	$x \neq 0$

(b) $|2x| - 3 = 0 \Leftrightarrow |x| = \frac{3}{2}$, giving $x = \pm\frac{3}{2}$. For $|2x| - 3 < 0$: $|x| < \frac{3}{2}$, hence $-\frac{3}{2} < x < \frac{3}{2}$. For $|2x| - 3 > 0$: $x \leftarrow \frac{3}{2}$ or $x > \frac{3}{2}$. The source answer's "all real" claim for $|x| > 0$ is corrected: zero is excluded.

NK2 Christos Nikolaidis · Math-AA 2.15 · original Q2 · source p.2

2. [Maximum mark: 8] ***[without GDC]***

(a) Solve the equation $|x-5|=3$. [3]

(b) Solve the inequalities (i) $|x-5|<3$ (ii) $|x-5|>3$ [5]

METHOD A: using the definition of $|x|$

METHOD B: by squaring both sides (since both sides are positive)

FF50 Further Functions · original Q50 · source p.52

50

No calculator **Haese AA HL Ex 6C.2 · Q1**

[Maximum mark: 9]

Solve for x:

(a) $|x|=3$ **(b)** $|x|=-5$ **(c)** $|x|=0$

(d) $|x-1|=3$ **(e)** $|3-x|=4$ **(f)** $|x+5|=-1$

(g) $|3x-2|=1$ **(h)** $|3-2x|=3$ **(i)** $|2-5x|=12$

FF53 Further Functions · original Q53 · source p.52

53

No calculator **Haese AA HL Ex 6C.2 · Q4**

[Maximum mark: 2]

Prove that if $|x|=|a|$ then $x=\pm a$.

NK2 Christos Nikolaidis · Math-AA 2.15 · original Q2 · source p.2

2. **METHOD A**

(a) $|x-5|=3 \Leftrightarrow x-5=\pm 3$

Hence, $x=8$, $x=2$

(b) $|x-5|<3 \Leftrightarrow -3<x-5<3$

Hence, $2<x<8$

(c) $|x-5|>3 \Leftrightarrow x-5<-3$ or $x-5>3 \Leftrightarrow x<2$ or $x>8$

Hence, $x<2$ or $x>8$

METHOD B

(a) $|x-5|=3 \Leftrightarrow |x-5|^2=3^2 \Leftrightarrow (x-5)^2=9 \Leftrightarrow x^2-10x+16=0$

$\Delta=36$, $x=\dfrac{10\pm 6}{2}$

Hence, $x=8$, $x=2$

(b) $|x-5|<3 \Leftrightarrow |x-5|^2<3^2 \Leftrightarrow (x-5)^2<9 \Leftrightarrow x^2-10x+16<0$

Hence, $2<x<8$

(c) $|x-5|>3 \Leftrightarrow |x-5|^2>3^2 \Leftrightarrow (x-5)^2>9 \Leftrightarrow x^2-10x+16>0$

Hence, $x<2$ or $x>8$

FF50 Further Functions · original Q50 · source p.52

50 *Haese AA HL Ex 6C.2 · Q1*

(a) $x=\pm 3$ **(b)** no solution **(c)** $x=0$

(d) $x=4$ or -2 **(e)** $x=-1$ or 7 **(f)** no solution

(g) $x=1$ or $\dfrac{1}{3}$ **(h)** $x=0$ or 3 **(i)** $x=-2$ or $\dfrac{14}{5}$

e.g. (i) $2-5x=\pm 12 \Rightarrow x=-2$ or $\frac{14}{5}$. (b), (f): a modulus is never negative.

FF53 Further Functions · original Q53 · source p.52

53 *Haese AA HL Ex 6C.2 · Q4*

$|x|=|a| \Rightarrow |x|^2=|a|^2 \Rightarrow x^2=a^2$ **M1**

$\Rightarrow x^2-a^2=0 \Rightarrow (x-a)(x+a)=0 \Rightarrow x=\pm a$ **A1**

NK9 Christos Nikolaidis · Math-AA 2.15 · original Q9 · source p.8

9. [Maximum mark: 6] ***[without GDC]***

(a) Solve the equation $\left|\frac{2x-6}{7}\right| = 8$. [3]

(b) Solve the inequality $\left|\frac{2x-6}{7}\right| \leq 8$. [3]

NK10 Christos Nikolaidis · Math-AA 2.15 · original Q10 · source p.8

10. [Maximum mark: 4] ***[without GDC]***

Solve the equation $|x^2 - 9| = 7$.

NK9

Christos Nikolaidis · Math-AA 2.15 · original Q9 · source p.8

9. (a) $\left|\frac{2x-6}{7}\right| = 8 \Leftrightarrow \frac{2x-6}{7} = \pm 8 \Leftrightarrow 2x-6 = \pm 56 \Leftrightarrow x-3 = \pm 28$

$x = 31$ or $x = -25$

(b) $\left|\frac{2x-6}{7}\right| \leq 8 \Leftrightarrow -8 \leq \frac{2x-6}{7} \leq 8 \Leftrightarrow -56 \leq 2x-6 \leq 56 \Leftrightarrow -50 \leq 2x \leq 62$

$-25 \leq x \leq 31$

NK10

Christos Nikolaidis · Math-AA 2.15 · original Q10 · source p.8

10. $|x^2-9| = 7 \Leftrightarrow x^2-9 = \pm 7 \Leftrightarrow x^2 = 16$ or $x^2 = 2$

$x = \pm 4$ or $x = \pm\sqrt{2}$

FF54 Further Functions · original Q54 · source p.54

54

No calculator **Haese AA HL Ex 6C.2 · Q5**

[Maximum mark: 6]

Solve for x:

(a) $|3x-1| = |x+2|$ **(b)** $|2x+5| = |1-x|$ **(c)** $|x+1| = |2-x|$

(d) $|x| = |5-x|$ **(e)** $|1-4x| = 2|x-1|$ **(f)** $|3x+2| = 2|2-x|$

FF55 Further Functions · original Q55 · source p.54

55

No calculator **Haese AA HL Ex 6C.2 · Q6**

[Maximum mark: 5]

(a) Sketch the graphs of $y = |x+3|$ and $y = 7 - \frac{x}{3}$ on the same set of axes. [2]

(b) Solve the equation $|x+3| = 7 - \frac{x}{3}$. [3]

NK4 Christos Nikolaidis · Math-AA 2.15 · original Q4 · source p.4

4. [Maximum mark: 8] ***[with / without GDC]***

(a) Solve the equation $|x-5| = |x-3|$ [4]

(b) Solve the inequality $|x-5| < |x-3|$ [4]

FF54 Further Functions · original Q54 · source p.54

54 *Haese AA HL Ex 6C.2 · Q5*

Use $|A| = |B| \Leftrightarrow A = \pm B$.

(a) $x = \frac{3}{2}$ or $-\frac{1}{4}$ **(b)** $x = -\frac{4}{3}$ or -6 **(c)** $x = \frac{1}{2}$

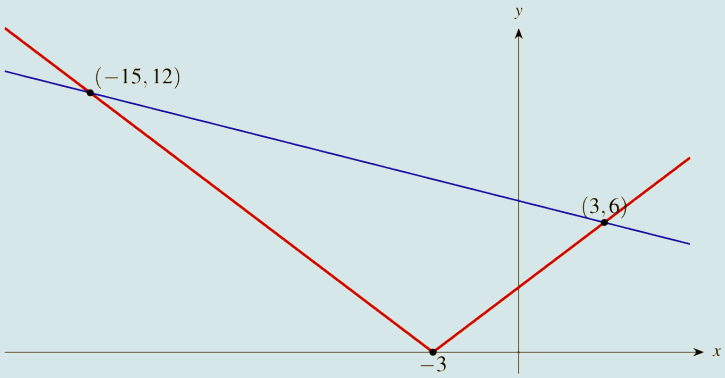
(d) $x = \frac{5}{2}$ **(e)** $x = \pm\frac{1}{2}$ **(f)** $x = \frac{2}{5}$ or -6

(c): $x+1 = -(2-x)$ gives $1 = -2$ — no solution; (d) similarly only one solution.

FF55 Further Functions · original Q55 · source p.54

55 *Haese AA HL Ex 6C.2 · Q6*

(a)



(b) $x \geq -3$: $x + 3 = 7 - \frac{x}{3} \Rightarrow \frac{4x}{3} = 4 \Rightarrow x = 3$

$x < -3$: $-x - 3 = 7 - \frac{x}{3} \Rightarrow -\frac{2x}{3} = 10 \Rightarrow x = -15$ $\boxed{x = -15 \text{ or } 3}$ **M1 A1 A1**

NK4 Christos Nikolaidis · Math-AA 2.15 · original Q4 · source p.4

4. With GDC

(a) **EITHER** SolveN **OR** graph

$x = 4$

(b) Graph of $y = |x-5| - |x-3|$, observe when $y < 0$

$x > 4$

Without GDC Square both sides (since both sides are positive)

(a) $(x-5)^2 = (x-3)^2 \Leftrightarrow x^2 - 10x + 25 = x^2 - 6x + 9 \Leftrightarrow 16 = 4x \Leftrightarrow x = 4$

(b) $(x-5)^2 < (x-3)^2 \Leftrightarrow x^2 - 10x + 25 < x^2 - 6x + 9 \Leftrightarrow 16 < 4x \Leftrightarrow x > 4$

NK6 Christos Nikolaidis · Math-AA 2.15 · original Q6 · source p.5

6*. [Maximum mark: 10] ***[with / without GDC]***

(a) Solve the equation $|x-5| = x-3$ [5]

(b) Solve the inequality $|x-5| < x-3$ [5]

NK6 Christos Nikolaidis · Math-AA 2.15 · original Q6 · source p.5

6. **With GDC**

(a) **EITHER** SolveN **OR** graph

$x = 4$

(b) Graph of $y = |x-5| - x + 3$, observe when $y < 0$

$x > 4$

Without GDC Cases

(a) **Case 1:** If $x \geq 5$,

The equation takes the form $x - 5 = x - 3 \Leftrightarrow 5 = 3$ which is impossible. No solutions!

Case 2: If $x < 5$,

The equation takes the form $5 - x = x - 3 \Leftrightarrow 8 = 2x \Leftrightarrow x = 4$ which is accepted.

Therefore, there is only one solution: $x = 4$

(b) **Case 1:** If $x \geq 5$,

The inequality takes the form $x - 5 < x - 3 \Leftrightarrow -5 < -3$ which is always true.

In this region, the whole region is accepted $x \geq 5$

Case 2: If $x < 5$,

The inequality takes the form $5 - x < x - 3 \Leftrightarrow 8 < 2x \Leftrightarrow x > 4$.

In this region, the solution is $4 < x < 5$

Therefore, the final solution is the union of the two sets: $x \in]4, +\infty[$

NK7 Christos Nikolaidis · Math-AA 2.15 · original Q7 · source p.6

7*. [Maximum mark: 12] ***[with / without GDC]***

(a) Solve the equation $|x-5|=|x|-3$ [6]

(b) Solve the inequality $|x-5|<|x|-3$ [6]

NK7 Christos Nikolaidis · Math-AA 2.15 · original Q7 · source p.6

7. **With GDC**

(a) **EITHER** SolveN **OR** graph

$x = 4$

(b) Graph of $y = |x-5| - x + 3$, observe when $y < 0$

$x > 4$

Without GDC Cases

The zeros of the moduli are $x = 0$ and $x = 5$. Hence, we consider three cases:

	0		**5**	
√		√		√

(a) **Case 1:** If $x \geq 5$,

The equation takes the form $x - 5 = x - 3 \Leftrightarrow 5 = 3$ which is impossible. No solutions!

Case 2: If $0 \leq x < 5$,

The equation takes the form $-x + 5 = x - 3 \Leftrightarrow 8 = 2x \Leftrightarrow x = 4$ which is accepted.

Case 3: If $x < 0$,

The equation takes the form $-x + 5 = -x - 3 \Leftrightarrow 5 = -3$ which is impossible.

Therefore, there is only one solution: $x = 4$

(b) **Case 1:** If $x \geq 5$,

The inequality takes the form $x - 5 < x - 3 \Leftrightarrow -5 < -3$ which is always true.

The whole region is accepted $x \geq 5$

Case 2: If $0 \leq x < 5$,

The inequality takes the form $-x + 5 < x - 3 \Leftrightarrow 8 < 2x \Leftrightarrow x > 4$.

In this region, the solution is $4 < x < 5$

Case 3: If $x < 0$,

The inequality takes the form $-x + 5 < -x - 3 \Leftrightarrow 5 < -3$ which is impossible.

The whole region is rejected.

Therefore, the final solution is: $x \in]4, +\infty[$

FF56 Further Functions · original Q56 · source p.56

56

Calculator

Haese AA HL Ex 6C.2 · Q7

[Maximum mark: 6]

Solve graphically:

(a) $|x+2| = 2x+1$ **(b)** $|2x+3| = 3|x| - 1$ **(c)** $|x-2| = \frac{2}{5}x + 1$

(d) $x^2 - 1 = |5x - x^2|$ **(e)** $|e^x - 6| = 2^x$ **(f)** $x^3 - 4x = \ln|x|$

FF56 Further Functions · original Q56 · source p.56

56 *Haese AA HL Ex 6C.2 · Q7*

(a) $\boxed{x = 1}$

(b) $\boxed{x = -0.8 \text{ or } 4}$

(c) $\boxed{x = \frac{5}{7} \approx 0.714 \text{ or } 5}$

(d) $\boxed{x \approx 2.69}$ $\left(= \frac{5+\sqrt{33}}{4}\right)$

(e) $\boxed{x \approx 1.28 \text{ or } 2.43}$

(f) $\boxed{x \approx -1.91,\ 0.304 \text{ or } 2.09}$

TI-Nspire CX: Question 56

home → **Add Graphs**. Type the function in the entry line, e.g. `f1(x)=abs(x+2)` (`abs` from the ctrl × template palette or typed), enter. Enter the second side as `f2(x)`. menu → **4: Window/-Zoom** to adjust. Then menu → **6: Analyze Graph** → **4: Intersection** (click left of and right of each crossing). e.g. (e) `f1(x)=abs(e^(x)-6)`, `f2(x)=2^x`: intersections at $x = 1.28,\ 2.43$. Check every crossing — use a wide window first.

FF51 Further Functions · original Q51 · source p.52

51

No calculator **Haese AA HL Ex 6C.2 · Q2**

[Maximum mark: 6]

Solve for x:

(a) $\left|\dfrac{x}{x-1}\right| = 3$ **(b)** $\left|\dfrac{2x-1}{x+1}\right| = 5$ **(c)** $\left|\dfrac{x+3}{1-3x}\right| = \dfrac{1}{2}$

FF52 Further Functions · original Q52 · source p.52

52

No calculator **Haese AA HL Ex 6C.2 · Q3**

[Maximum mark: 3]

(a) Explain why the equation $\left|\dfrac{3x+1}{x-1}\right| = 3$ has only one solution. [2]

(b) Find the solution. [1]

NK13 Christos Nikolaidis · Math-AA 2.15 · original Q13 · source p.10

13. [Maximum mark: 5] ***[with GDC]***

Solve the inequality $\left|\dfrac{x+9}{x-9}\right| \leq 2$.

FF51 Further Functions · original Q51 · source p.52

51 *Haese AA HL Ex 6C.2 · Q2*

(a) $\frac{x}{x-1} = \pm 3$: $x = 3x-3 \Rightarrow x = \frac{3}{2}$; $x = -3x+3 \Rightarrow x = \frac{3}{4}$. $\boxed{x = \frac{3}{4} \text{ or } \frac{3}{2}}$

(b) $2x-1 = \pm 5(x+1)$: $x = -2$ or $7x = -4$: $\boxed{x = -2 \text{ or } -\frac{4}{7}}$

(c) $2(x+3) = \pm(1-3x)$: $5x = -5$ or $-x = 7$: $\boxed{x = -1 \text{ or } 7}$

FF52 Further Functions · original Q52 · source p.52

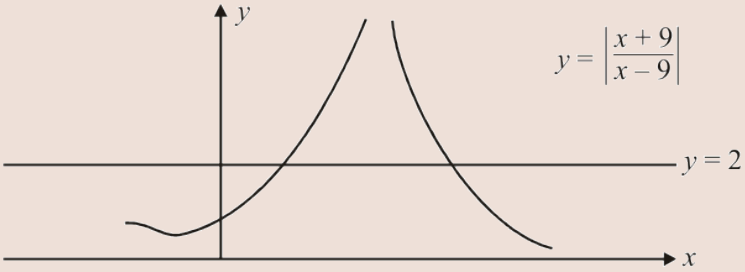
52 *Haese AA HL Ex 6C.2 · Q3*

(a) $\frac{3x+1}{x-1} = 3 \Rightarrow 3x+1 = 3x-3 \Rightarrow 1 = -3$, impossible (the graph never reaches its H.A. $y = 3$), so only $\frac{3x+1}{x-1} = -3$ gives a solution. **R2**

(b) $3x+1 = -3x+3 \Rightarrow \boxed{x = \frac{1}{3}}$ **A1**

NK13 Christos Nikolaidis · Math-AA 2.15 · original Q13 · source p.10

13. **METHOD 1** (with GDC)



$]-\infty, 3] \cup [27, \infty[$

METHOD 2 (taking squares)

$|x+9| \le 2|x-9|$ and then taking squares:

$x^2 + 18x + 81 \le 4(x^2 - 18x + 81)$

$3x^2 - 5\times 18x + 3\times 81 \ge 0 \Leftrightarrow x^2 - 30x + 81 \ge 0 \Leftrightarrow (x-3)(x-27) \ge 0$

$x \in]-\infty, 3] \cup [27, \infty[$

METHOD 3 (solving analytically; not recommended, just as an option)

Solve $-2 \le \frac{x+9}{x-9} \le 2$ (two relations)

$$\frac{x+9}{x-9} \le 2 \Leftrightarrow \frac{x+9}{x-9} - 2 \le 0 \Leftrightarrow \frac{-x+27}{x-9} \le 0 \Leftrightarrow \frac{x-27}{x-9} \ge 0 \quad \text{so } x < 9 \text{ or } x \ge 27$$

$$\frac{x+9}{x-9} \ge -2 \Leftrightarrow \frac{x+9}{x-9} + 2 \ge 0 \Leftrightarrow \frac{3x-9}{x-9} \ge 0 \Leftrightarrow \frac{x-3}{x-9} \ge 0 \quad \text{so } x \le 3 \text{ or } x > 9$$

Therefore, $x \in]-\infty, 3] \cup [27, \infty[$

NK14 Christos Nikolaidis · Math-AA 2.15 · original Q14 · source p.10

14. [Maximum mark: 6] ***[without GDC]***

Solve the inequality $\left|\frac{x+1}{x}\right| \geq 2$.

FF59 Further Functions · original Q59 · source p.58

59

No calculator **Haese AA HL Ex 6C.3 · Q3**

[Maximum mark: 6]

Solve for x:

(a) $\left|\frac{x}{x-2}\right| \geq 3$ **(b)** $\left|\frac{2x+3}{x-1}\right| \geq 2$ **(c)** $\left|\frac{x-4}{1-2x}\right| < \frac{2}{3}$

FF66 Further Functions · original Q66 · source p.62

66

Calculator **Haese Review 6B · Q10**

[Maximum mark: 4]

Solve graphically:

(a) $\left|\frac{4}{x-1}\right| = 6 - \frac{x}{2}$ **(b)** $\frac{x}{|x|+1} \geq \frac{1}{3}$

NK14 Christos Nikolaidis · Math-AA 2.15 · original Q14 · source p.10

14. $\left|\frac{x+1}{x}\right| \geq 2 \Leftrightarrow |x+1| \geq 2|x| \Leftrightarrow |x+1|^2 \geq 4|x|^2 \Leftrightarrow (x+1)^2 \geq 4x^2$

$\Leftrightarrow x^2+2x+1 \geq 4x^2 \Leftrightarrow 3x^2-2x-1 \leq 0$

$\Delta = 16$, roots $x = \frac{2 \pm 4}{6} = 1, \ -\frac{1}{3}$. Solution: $-\frac{1}{3} \leq x \leq 1$

FF59 Further Functions · original Q59 · source p.58

59 *Haese AA HL Ex 6C.3 · Q3*

(a) $x^2 \geq 9(x-2)^2, \ x \neq 2 \Rightarrow (2x-3)(x-3) \leq 0$: $\boxed{\frac{3}{2} \leq x \leq 3, \ x \neq 2}$

(b) $(2x+3)^2 \geq 4(x-1)^2, \ x \neq 1 \Rightarrow 20x \geq -5$: $\boxed{x \geq -\frac{1}{4}, \ x \neq 1}$

(c) $9(x-4)^2 < 4(1-2x)^2 \Rightarrow x^2+8x-20 > 0 \Rightarrow (x+10)(x-2) > 0$: $\boxed{x < -10 \text{ or } x > 2}$

FF66 Further Functions · original Q66 · source p.62

66 *Haese Review 6B · Q10*

(a) $\boxed{x \approx 0.315, \ 1.78 \text{ or } 11.2}$

(b) intersection at $x = \frac{1}{2}$ (LHS < 0 for $x < 0$): $\boxed{x \geq \frac{1}{2}}$

TI-Nspire CX: Question 66

home → **Add Graphs**. Type the function in the entry line, e.g. `f1(x)=abs(x+2)` (abs from the ctrl × template palette or typed), enter. Enter the second side as `f2(x)`. menu → **4: Window/-Zoom** to adjust. Then menu → **6: Analyze Graph** → **4: Intersection** (click left of and right of each crossing). (a) `f1(x)=abs(4/(x-1))`, `f2(x)=6-x/2`; use window $-2 \leq x \leq 14$ to see all three intersections.

FF48 Further Functions · original Q48 · source p.48

48

No calculator **RV · Q7**

[Maximum mark: 8]

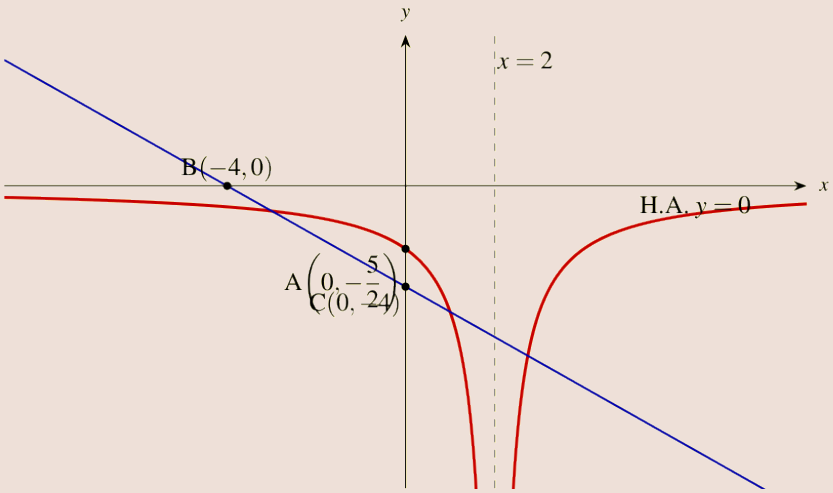
(a) Sketch the curve $y = -\left|\frac{5}{x-2}\right|$ and line $y = -x-4$ on the same axes, clearly indicating any x and y intercepts and any asymptotes. [3]

(b) Find the exact solutions to the equation $x+4 = \frac{5}{|x-2|}$. [5]

FF48 Further Functions · original Q48 · source p.48

48 *RV · Q7*

(a) **A1 A1 A1**



(b) Equivalent to $-x-4=-\left|\dfrac{5}{x-2}\right|$ (intersections of the graphs).

$x<2$: $(x+4)(x-2)=-5 \Rightarrow x^2+2x-3=0 \Rightarrow (x+3)(x-1)=0$: $x=-3,\ 1$ **M1**

A1

$x>2$: $(x+4)(x-2)=5 \Rightarrow x^2+2x-13=0 \Rightarrow x=-1\pm\sqrt{14}$; need $x>2$: $x=-1+\sqrt{14}$ **M1 A1**

$\boxed{x=-3,\ x=1,\ x=-1+\sqrt{14}}$ **A1**

FF49 Further Functions · original Q49 · source p.50

49

No calculator **RV · Q8**

[Maximum mark: 17]

The function f is defined by $f(x) = 1 + \dfrac{4x}{x+3}$, for $x \neq -3$.

(a) Sketch the graph of $y = f(x)$, indicating clearly any asymptotes and points of intersection with the x and y axes. [4]

(b) Find an expression for f^{-1}. [4]

(c) Find all values of x for which $f(x) = f^{-1}(x)$. [3]

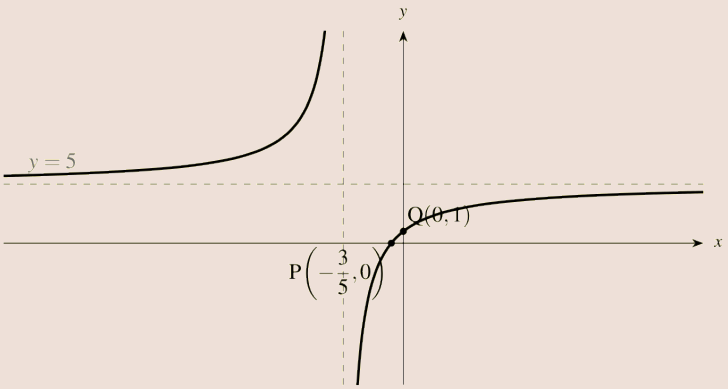
(d) Solve the inequality $|f(x)| < 2$. [4]

(e) Solve the inequality $f(|x|) < 2$. [2]

FF49 Further Functions · original Q49 · source p.50

49 *RV · Q8*

(a) $f(x)=\dfrac{5x+3}{x+3}$: x-int $-\dfrac{3}{5}$, y-int 1, V.A. $x=-3$, H.A. $y=5$ **A1 A1 A1 A1**



(b) $x=\dfrac{5y+3}{y+3} \Rightarrow xy+3x=5y+3 \Rightarrow y(x-5)=3-3x$ **M1 M1 M1**

$\boxed{f^{-1}(x)=\dfrac{3x-3}{5-x}}$ **A1**

(c) $\dfrac{5x+3}{x+3}=\dfrac{3x-3}{5-x} \Rightarrow (5x+3)(5-x)=(3x-3)(x+3)$ **M1**

$\Rightarrow -8x^2+16x+24=0 \Rightarrow x^2-2x-3=0 \Rightarrow \boxed{x=-1,\ 3}$ **A1 A1**

(d) $-2<f(x)<2$; f is increasing on each branch. $f(x)=2 \Rightarrow x=f^{-1}(2)=1$;

$f(x)=-2 \Rightarrow x=f^{-1}(-2)=-\dfrac{9}{7}$ **M1 A1 A1**

(on the left branch $f>5$, so no solutions there): $\boxed{-\dfrac{9}{7}<x<1}$ **A1**

(e) $|x|\geq 0$ so $f(|x|)\geq 1>-2$; $f(|x|)<2 \Leftrightarrow |x|<1$: $\boxed{-1<x<1}$ **M1 A1**

FF57 Further Functions · original Q57 · source p.56

57

No calculator **Haese AA HL Ex 6C.3 · Q1**

[Maximum mark: 6]

Solve for x:

(a) $|x-1| \leq 4$ **(b)** $|x+2| > 7$ **(c)** $|2x-3| < 1$

(d) $|3x+5| \geq 2$ **(e)** $|1-2x| < 4$ **(f)** $|-5x-4| > \frac{1}{2}$

FF58 Further Functions · original Q58 · source p.56

58

No calculator **Haese AA HL Ex 6C.3 · Q2**

[Maximum mark: 6]

Solve for x:

(a) $|x+5| < |x-1|$ **(b)** $|2x+1| > |x+2|$ **(c)** $|x-3| \leq |2x-6|$

(d) $|3x+4| \geq |x+3|$ **(e)** $2|x-4| < |3x+2|$ **(f)** $\frac{1}{3}|2x+5| \geq |4-x|$

NK5 Christos Nikolaidis · Math-AA 2.15 · original Q5 · source p.4

5*. [Maximum mark: 5] ***[with / without GDC]***

Solve the inequality $x^2 - 3|x| + 2 > 0$

FF57 Further Functions · original Q57 · source p.56

57 *Haese AA HL Ex 6C.3 · Q1*

$|A| < k \Leftrightarrow -k < A < k$; $|A| > k \Leftrightarrow A < -k$ or $A > k$.

(a) $-3 \leq x \leq 5$ **(b)** $x < -9$ or $x > 5$ **(c)** $1 < x < 2$

(d) $x \leq -\frac{7}{3}$ or $x \geq -1$ **(e)** $-\frac{3}{2} < x < \frac{5}{2}$ **(f)** $x < -\frac{9}{10}$ or $x > -\frac{7}{10}$

FF58 Further Functions · original Q58 · source p.56

58 *Haese AA HL Ex 6C.3 · Q2*

Square both sides (both sides ≥ 0):

(a) $x < -2$ **(b)** $x < -1$ or $x > 1$ **(c)** $x \in \mathbb{R}$

(d) $x \leq -\frac{7}{4}$ or $x \geq -\frac{1}{2}$ **(e)** $x < -10$ or $x > \frac{6}{5}$ **(f)** $\frac{7}{5} \leq x \leq 17$

e.g. (d) $(3x+4)^2 \geq (x+3)^2 \Rightarrow 8x^2+18x+7 \geq 0 \Rightarrow (4x+7)(2x+1) \geq 0$; (c) $3(x-3)^2 \geq 0$ is always true.

NK5 Christos Nikolaidis · Math-AA 2.15 · original Q5 · source p.4

5. **With GDC** Graph of $y = x^2 - 3|x| + 2$, observe when $y > 0$

$x \in]-\infty, -2[\cup]-1, 1[\cup]2, +\infty[$

Without GDC

METHOD A: Cases

Case 1: If $x \geq 0$, the equation takes the form $x^2 - 3x + 2 > 0$

Roots 1, 2, $x < 1$ or $x > 2$

$x \in [0, 1[\cup]2, +\infty[$

Case 2: If $x < 0$, the equation takes the form $x^2 + 3x + 2 > 0$

Roots –1, –2, $x < -2$ or $x > -1$

$x \in]-\infty, -2[\cup]-1, 0[$

Therefore, the final solution is the union of the two sets: $x \in]-\infty, -2[\cup]-1, 1[\cup]2, +\infty[$

METHOD B: it is a hidden quadratic,

Let $y = |x|$. The inequality takes the form: $y^2 - 3y + 2 > 0 \Leftrightarrow y < 1$ or $y > 2$

$|x| < 1$ gives $-1 < x < 1$, $|x| > 2$ gives $x < -2$ or $x > 2$

Final solution: $x \in]-\infty, -2[\cup]-1, 1[\cup]2, +\infty[$

NK11 Christos Nikolaidis · Math-AA 2.15 · original Q11 · source p.9

11. [Maximum mark: 5] ***[with / without GDC]***

Find the values of x for which $|5-3x| \leq |x+1|$.

NK12 Christos Nikolaidis · Math-AA 2.15 · original Q12 · source p.9

12. [Maximum mark: 5] ***[with / without GDC]***

Solve the inequality $|x-2| \geq |2x+1|$.

NK15 Christos Nikolaidis · Math-AA 2.15 · original Q15 · source p.11

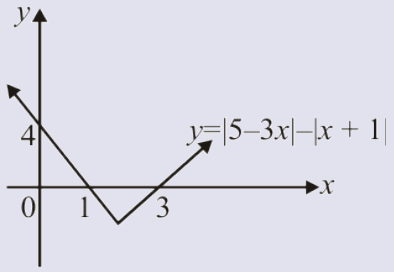
15. [Maximum mark: 6] ***[without GDC]***

Solve the inequalities (i) $|x-5|+|x-10| \leq 7$. (ii) $|x-5|-|x-10| \leq 7$

Notice: confirm the result by using the graph mode of the GDC

NK11 Christos Nikolaidis · Math-AA 2.15 · original Q11 · source p.9

11. **With GDC** Sketch the graph of $y = |5 - 3x| - |x + 1|$.



From this graph we see that $y \le 0$ for $1 \le x \le 3$.

Without GDC (take squares)

$|5 - 3x| \le |x + 1| \Rightarrow 25 - 30x + 9x^2 \le x^2 + 2x + 1$
$\Rightarrow 8x^2 - 32x + 24 \le 0 \Rightarrow 8(x - 1)(x - 3) \le 0$
$\Rightarrow 1 \le x \le 3$

NK12 Christos Nikolaidis · Math-AA 2.15 · original Q12 · source p.9

12. **METHOD 1** (without GDC, taking squares)

$(x - 2)^2 \ge (2x + 1)^2 \Leftrightarrow x^2 - 4x + 4 \ge 4x^2 + 4x + 1$

$3x^2 + 8x - 3 \le 0 \Leftrightarrow (3x - 1)(x + 3) \le 0$ (or find roots of equation)

So $x \in \left[-3, \frac{1}{3}\right]$.

METHOD 2 (with GDC)

Observe the graph of $y = |x - 2| - |2x + 1|$ and check when $y \ge 0$: $x \in \left[-3, \frac{1}{3}\right]$

NK15 Christos Nikolaidis · Math-AA 2.15 · original Q15 · source p.11

15. **(i)** $|x-5| + |x-10| \le 7$

zeros of abs. values: 5, 10

Within $x \ge 10$: $(x-5) + (x-10) \le 7 \Leftrightarrow 2x \le 22 \Leftrightarrow x \le 11$ $\quad 10 \le x \le 11$

Within $5 \le x < 10$: $(x-5) - (x-10) \le 7 \Leftrightarrow 5 \le 7$, true $\quad 5 \le x < 10$

Within $x < 5$: $-(x-5) - (x-10) \le 7 \Leftrightarrow -2x \le -8 \Leftrightarrow x \ge 4$, true $\quad 4 \le x < 5$

The solution set is the union of the 3 intervals: $4 \le x \le 11$

(ii) $|x-5| - |x-10| \le 7$

A similar analysis shows that the result is true for any x. The solution set is $x \in \mathbb{R}$

FF60 Further Functions · original Q60 · source p.58

60

Calculator **Haese AA HL Ex 6C.3 · Q4**

[Maximum mark: 8]

Solve graphically:

(a) $|2x-3| < x$

(b) $|x|-2 \geq |4-x|$

(c) $x^2-4 > |x-1|$

(d) $|2x-1|+|x-4| \leq 10-x$

(e) $\ln|x^2-x+5| \geq 0.1x^4$

(f) $7|x|-e^{|x|} \geq \log_2(x^2-x-2)$

(g) $|3x\arccos x| > 1$

(h) $0.2\ln(7-2x) < |\sqrt{15-x^2}-3|$

FF61 Further Functions · original Q61 · source p.60

61

No calculator **Haese Review 6A · Q10**

[Maximum mark: 4]

Solve for x:

(a) $\left|\dfrac{2x+1}{x-2}\right| = 3$

(b) $|3x+5| = |x-7|$

FF62 Further Functions · original Q62 · source p.60

62

No calculator **Haese Review 6A · Q11**

[Maximum mark: 6]

Solve for x:

(a) $|x+3| \leq 6$

(b) $|3-2x| > 1$

(c) $|3x+2| < |x-5|$

FF60 Further Functions · original Q60 · source p.58

60 *Haese AA HL Ex 6C.3 · Q4*

(a) $1 < x < 3$

(b) $x \geq 3$

(c) $x < \dfrac{-1-\sqrt{21}}{2} \approx -2.79$ or $x > \dfrac{1+\sqrt{13}}{2} \approx 2.30$

(d) $-\dfrac{5}{2} \leq x \leq \dfrac{7}{2}$

(e) $-2.24 \leq x \leq 2.11$

(f) $-2.80 \leq x < -1$ or $2 < x \leq 2.92$ (domain $x < -1$ or $x > 2$)

(g) $-1 \leq x < -0.189$ or $0.254 < x < 0.937$ (domain $-1 \leq x \leq 1$)

(h) $-\sqrt{15} \leq x < -2.97$, $-1.72 < x < 2.19$ or $2.59 < x < 3.5$

TI-Nspire CX: Question 60

home → **Add Graphs**. Type the function in the entry line, e.g. `f1(x)=abs(x+2)` (`abs` from the ctrl × template palette or typed), enter. Enter the second side as `f2(x)`. menu → **4: Window/-Zoom** to adjust. Then menu → **6: Analyze Graph** → **4: Intersection** (click left of and right of each crossing). Graph LHS as f1 and RHS as f2, find all intersections, then read off where f1 is above (or below) f2. Watch the **domain** (e.g. $\log_2$ needs $x^2 - x - 2 > 0$; arccos needs $-1 \leq x \leq 1$).

FF61 Further Functions · original Q61 · source p.60

61 *Haese Review 6A · Q10*

(a) $2x+1 = \pm 3(x-2)$: $x = 7$ or $5x = 5$: $\boxed{x = 1 \text{ or } 7}$

(b) $3x+5 = \pm(x-7)$: $2x = -12$ or $4x = 2$: $\boxed{x = -6 \text{ or } \dfrac{1}{2}}$

FF62 Further Functions · original Q62 · source p.60

62 *Haese Review 6A · Q11*

(a) $-6 \leq x+3 \leq 6$: $\boxed{-9 \leq x \leq 3}$

(b) $3-2x > 1$ or $3-2x < -1$: $\boxed{x < 1 \text{ or } x > 2}$

(c) $(3x+2)^2 < (x-5)^2 \Rightarrow 8x^2+22x-21 < 0 \Rightarrow (4x-3)(2x+7) < 0$: $\boxed{-\dfrac{7}{2} < x < \dfrac{3}{4}}$

FF63 Further Functions · original Q63 · source p.60

63

Calculator **Haese Review 6A · Q12**

[Maximum mark: 4]

Solve graphically:

(a) $|2x-6| > x+3$ **(b)** $\sqrt{x^2+x+5} \leq |7-2^x|$

FF64 Further Functions · original Q64 · source p.60

64

No calculator **Haese Review 6B · Q8**

[Maximum mark: 4]

Solve for x:

(a) $|2-5x| = 6$ **(b)** $|4x-2| = |x+7|$

FF63 Further Functions · original Q63 · source p.60

(a) Solve $|2x-6| > x+3$ by cases. For $x < 3$: $6-2x > x+3$, so $x < 1$. For $x \geq 3$: $2x-6 > x+3$, so $x > 9$. Therefore $\boldsymbol{x < 1}$ **or** $\boldsymbol{x > 9}$. A GDC graph of $|2x-6|$ and $x+3$ gives the same boundary values 1 and 9. (b) The printed inequality is $\sqrt{x^2+x+5} \leq |7-2x|$. Since $x^2+x+5 = \left(x+\frac{1}{2}\right)^2 + \frac{19}{4} > 0$, both sides are non-negative and squaring is equivalent:

$$x^2+x+5 \leq (7-2x)^2 \Leftrightarrow 3x^2-29x+44 \geq 0.$$

The roots are $x = \frac{29 \pm \sqrt{313}}{6}$. The quadratic opens upwards, so

$$x \leq \frac{29-\sqrt{313}}{6} \quad \text{or} \quad x \geq \frac{29+\sqrt{313}}{6}.$$

Numerically: $\boldsymbol{x \leq 1.8847}$ **or** $\boldsymbol{x \geq 7.7820}$. GDC: enter $\sqrt{x^2+x+5}$ and $|7-2x|$, **not** $|7-2^x|$. The latter erroneous calculator entry caused the incorrect source answer.

FF64 Further Functions · original Q64 · source p.60

64 *Haese Review 6B · Q8*

(a) $2-5x = \pm 6$: $\boxed{x = -\frac{4}{5} \text{ or } \frac{8}{5}}$

(b) $4x-2 = \pm(x+7)$: $\boxed{x = 3 \text{ or } -1}$

FF65 Further Functions · original Q65 · source p.62

65

No calculator **Haese Review 6B · Q9**

[Maximum mark: 7]

Solve for x:

(a) $|3x+2| > 3$ **(b)** $\frac{1}{4}|2x+5| \geq |x-1|$ **(c)** $\left|\frac{x}{x+4}\right| \leq 2$

NK18 Christos Nikolaidis · Math-AA 2.15 · original Q18 · source p.16

18. [Maximum mark: 12] ***[without GDC]***

Let $f(x) = |x|$, $g(x) = |2x-5|$ and $h(x) = \frac{x+4}{2}$

(a) Solve $f(x) > g(x)$ [5]

(b) Solve $f(x) < h(x)$ [5]

(c) **Hence**, solve $g(x) < f(x) < h(x)$. [2]

FF65 Further Functions · original Q65 · source p.62

65 *Haese Review 6B · Q9*

(a) $3x+2>3$ or $3x+2<-3$: $\boxed{x<-\frac{5}{3} \text{ or } x>\frac{1}{3}}$

(b) $(2x+5)^2 \geq 16(x-1)^2 \Rightarrow 12x^2-52x-9\leq 0 \Rightarrow (6x+1)(2x-9)\leq 0$:

$\boxed{-\frac{1}{6}\leq x\leq \frac{9}{2}}$

(c) $x^2 \leq 4(x+4)^2,\ x\neq -4 \Rightarrow 3x^2+32x+64\geq 0 \Rightarrow (3x+8)(x+8)\geq 0$:

$\boxed{x\leq -8 \text{ or } x\geq -\frac{8}{3}}$

NK18 Christos Nikolaidis · Math-AA 2.15 · original Q18 · source p.16

18. (a) $|x|>|2x-5| \Leftrightarrow |x|^2>|2x-5|^2 \Leftrightarrow x^2>(2x-5)^2 \Leftrightarrow x^2>4x^2-20x+25$

$\Leftrightarrow 3x^2-20x+25<0$

$\Delta = 400-300=100,$

$x=\frac{20\pm 10}{6}=\frac{5}{3}$ or 5

$\frac{5}{3}<x<5$

(b) $|x|<\frac{x+4}{2}$

If $x\geq 0$: $x<\frac{x+4}{2} \Leftrightarrow 2x<x+4 \Leftrightarrow x<4$ Hence $0\leq x<4$

If $x<0$: $-x<\frac{x+4}{2} \Leftrightarrow -2x<x+4 \Leftrightarrow 3x>-4 \Leftrightarrow x>-\frac{4}{3}$ Hence $-\frac{4}{3}<x<0$

Therefore, $-\frac{4}{3}<x<4$

(c) The intersection of (a) and (b): $\frac{5}{3}<x<4$

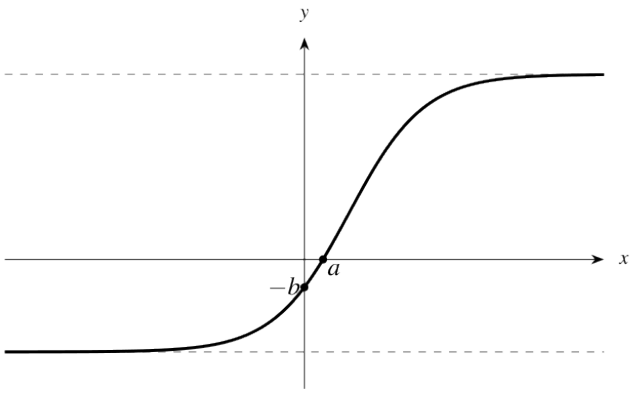
FF67 Further Functions · original Q67 · source p.64

67

No calculator RV · Q9

[Maximum mark: 6]

Part of the graph of a function, f, is shown in the following diagram. The graph of $y = f(x)$ has two horizontal asymptotes, $y = 4$ and $y = -2$. The x-intercept of $f(x)$ is a and the y-intercept is $-b$.



Consider the function $g(x) = |f(|x|)|$.

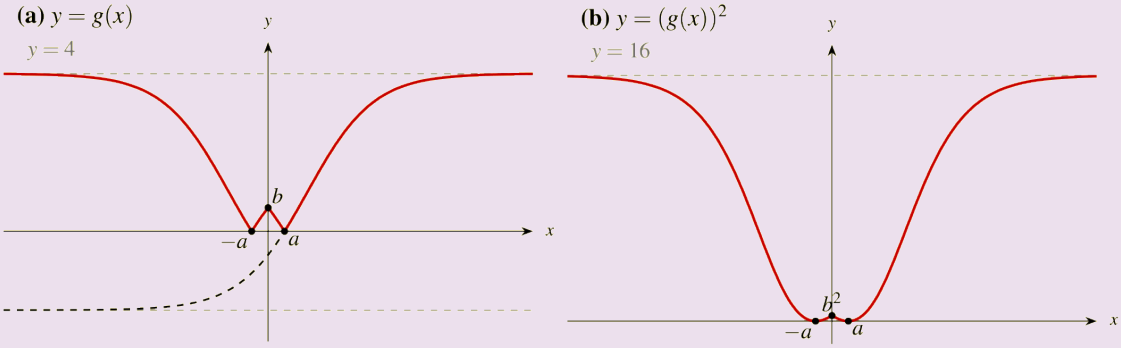
(a) Sketch the graph of $y = g(x)$ on top of the image above, labelling any new axis intercepts, in terms of a and b, and giving the equation of any new asymptotes. [3]

(b) Find the possible values of r such that $(g(x))^2 = r$ has exactly two solutions. [3]

FF67 Further Functions · original Q67 · source p.64

67 *RV · Q9*

(a) $f(|x|)$: keep $x \geq 0$, reflect in the y-axis; then $|\ |$ reflects the negative part up. New x-intercept $-a$, y-intercept b; the asymptote $y = 4$ applies on both sides. **A1 A1 A1**



(b) $(g(x))^2$ has y-intercept b^2 (a local maximum), touches the axis at $\pm a$, and H.A. $y = 16$. **M1**

A horizontal line $y = r$ cuts it exactly twice when $\boxed{b^2 < r < 16}$ or $\boxed{r = 0}$

A1 A1

Using the shape shown in the given sketch: g has both x-intercepts $x = \pm a$ and y-intercept b. Its square has asymptote $y = 16$ and y-intercept b^2. The stated intersection count $r = 0$ or $b^2 < r < 16$ follows from that displayed single-well shape, not from asymptote values alone.

FF68 Further Functions · original Q68 · source p.66

68

Calculator **Haese Investigation 1**

[Maximum mark: 6]

1. Sketch each pair of functions on the same set of axes. Include all axes intercepts and asymptotes.

(a) $y = x + 1$ and $y = (x+1)^2$ **(b)** $y = x^2 - 4$ and $y = (x^2-4)^2$

(c) $y = \dfrac{2}{x}$ and $y = \left(\dfrac{2}{x}\right)^2$ **(d)** $y = \dfrac{1}{x-3}$ and $y = \dfrac{1}{(x-3)^2}$

(e) $y = 2^x - 1$ and $y = (2^x - 1)^2$ **(f)** $y = \log x$ and $y = (\log x)^2$

2. Determine the points where each pair of functions intersect. What do you notice?

3. What other observations can you make about the graph of $y = [f(x)]^2$?

FF69 Further Functions · original Q69 · source p.68

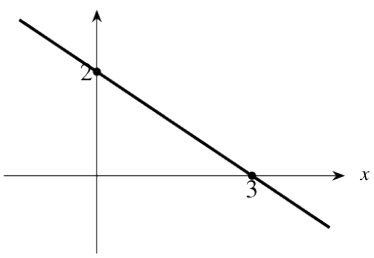
69

No calculator **Haese AA HL Ex 6B · Q1**

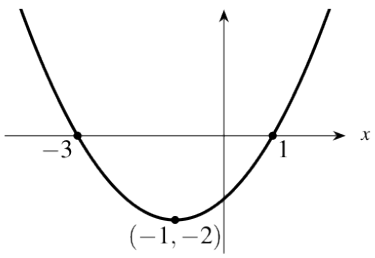
[Maximum mark: 6]

For each graph of $y = f(x)$, sketch the graph of $y = [f(x)]^2$.

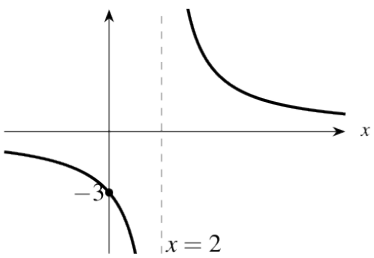
(a)



(b)



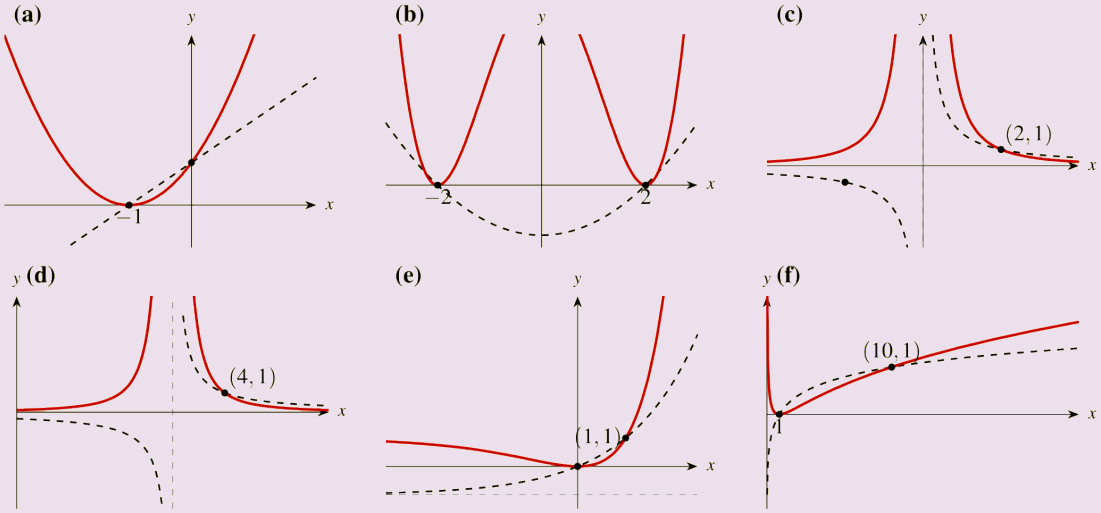
(c)



FF68 Further Functions · original Q68 · source p.66

68 *Haese Investigation 1*

(dashed: $y = f(x)$; red: new graph)



2. Intersections: (a) $(-1,0),(0,1)$ (b) $(\pm 2,0),(\pm\sqrt{5},1)$ (c) $(2,1)$ (d) $(4,1)$ (e) $(0,0),(1,1)$ (f) $(1,0),(10,1)$. They meet exactly where $f(x) = 0$ or $f(x) = 1$.

3. $[f(x)]^2 \geq 0$; it *touches* the x-axis at the zeros of f; vertical asymptotes stay; H.A. $y = k$ becomes $y = k^2$; values between -1 and 1 move closer to the axis, others further away.

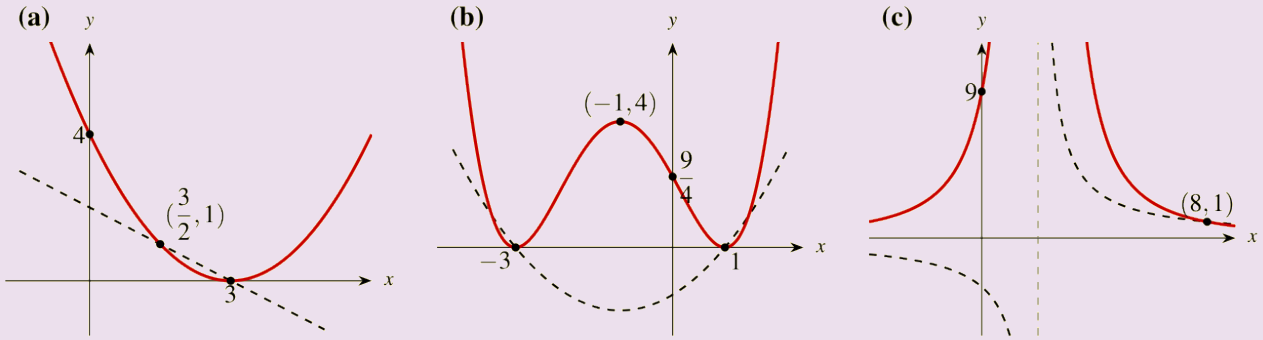
TI-Nspire CX: Question 68

Graphs: `f1(x)=x^2-4`, `f2(x)=(f1(x))^2`. menu → **6: Analyze Graph** → **4: Intersection**.

FF69 Further Functions · original Q69 · source p.68

69 *Haese AA HL Ex 6B · Q1*

(dashed: $y = f(x)$; red: new graph)



Key points: zeros of f become points where the graph *touches* the x-axis; points with $y = 1$ are invariant; turning point $(-1,-2) \to (-1,4)$; V.A. $x = 2$ remains. **A2 A2 A2**

FF70 Further Functions · original Q70 · source p.68

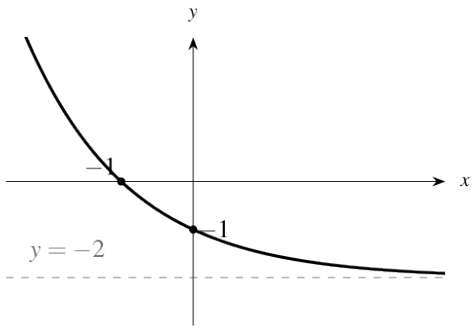
70

No calculator

Haese AA HL Ex 6B · Q2

[Maximum mark: 4]

Copy the graph alongside, and on the same set of axes, sketch the graph of $y = [f(x)]^2 + 1$.



FF71 Further Functions · original Q71 · source p.70

71

No calculator

Haese AA HL Ex 6B · Q3

[Maximum mark: 4]

(a) Draw the graph of $y = f(x) = 3^x$. [1]

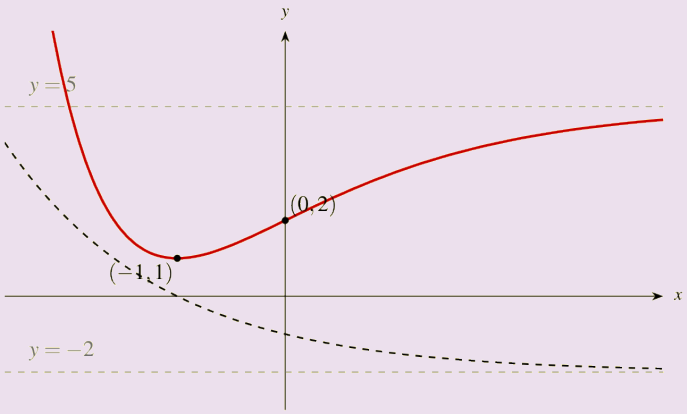
(b) On the same set of axes, sketch the graph of $y = [f(x)]^2$. [1]

(c) Describe a stretch which transforms $y = f(x)$ to $y = [f(x)]^2$. [2]

FF70 Further Functions · original Q70 · source p.68

70 *Haese AA HL Ex 6B · Q2*

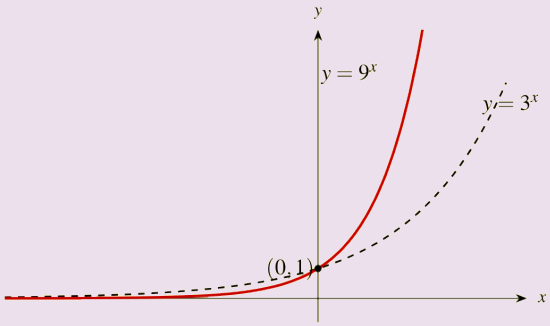
(dashed: $y = f(x)$; red: new graph)



$[f(x)]^2$ touches the axis at $(-1, 0)$ and has H.A. $y = (-2)^2 = 4$; adding 1 shifts up: minimum $(-1, 1)$, y-intercept $(-1)^2 + 1 = 2$, H.A. $\boxed{y = 5}$. **A1 A1 A1 A1**

FF71 Further Functions · original Q71 · source p.70

71 *Haese AA HL Ex 6B · Q3*



(b) $[f(x)]^2 = 3^{2x} = 9^x$: same H.A. $y = 0$, same point $(0, 1)$, steeper. **A1**

(c) $3^{2x} = f(2x)$: a **horizontal stretch with scale factor** $\frac{1}{2}$ (parallel to the x-axis). **A2**

FF72 Further Functions · original Q72 · source p.70

72

No calculator **Haese AA HL Ex 6B · Q4**

[Maximum mark: 4]

(a) Draw the graph of $y = f(x) = \log_3 x$. [1]

(b) On the same set of axes, sketch the graph of $y = [f(x)]^2$. [1]

(c) Find the invariant points when $y = f(x)$ is transformed to $y = [f(x)]^2$. [2]

FF73 Further Functions · original Q73 · source p.70

73

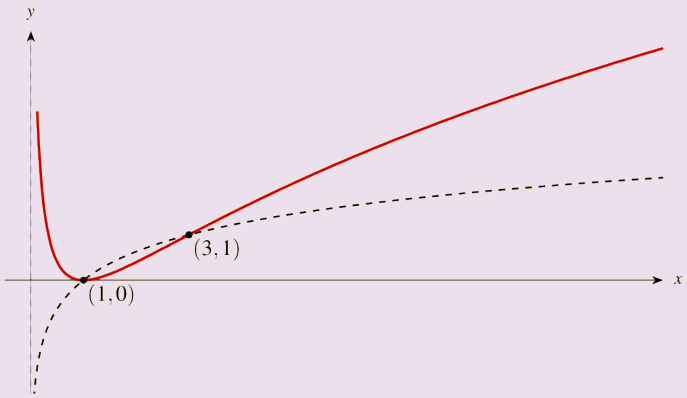
No calculator **Haese AA HL Ex 6B · Q6**

[Maximum mark: 2]

Suppose $f(x)$ has domain $0 \leq x \leq 5$ and range $-4 \leq y \leq 3$. Find the domain and range of $[f(x)]^2$.

FF72 Further Functions · original Q72 · source p.70

72 *Haese AA HL Ex 6B · Q4*



$[f(x)]^2$ touches the x-axis at $x = 1$; as $x \to 0^+$, $[f(x)]^2 \to +\infty$ (V.A. $x = 0$ remains). **A2**

(c) $f(x) = 0$ or 1: $\boxed{(1, 0) \text{ and } (3, 1)}$ **A1 A1**

FF73 Further Functions · original Q73 · source p.70

73 *Haese AA HL Ex 6B · Q6*

Domain $\boxed{0 \leq x \leq 5}$ (unchanged) **A1**

Range: $y = -4$ gives 16, $y = 0$ gives 0: $\boxed{0 \leq y \leq 16}$ **A1**

FF74 Further Functions · original Q74 · source p.72

74

No calculator **Haese AA HL Ex 6B · Q7**

[Maximum mark: 8]

Consider the function $f(x) = \dfrac{2x+4}{x-1}$.

(a) Find the axes intercepts and asymptotes of the function. [2]

(b) Hence find the axes intercepts and asymptotes of $y = [f(x)]^2$. [2]

(c) Which points are invariant when $y = f(x)$ is transformed to $y = [f(x)]^2$? [2]

(d) Sketch $y = f(x)$ and $y = [f(x)]^2$ on the same set of axes. [2]

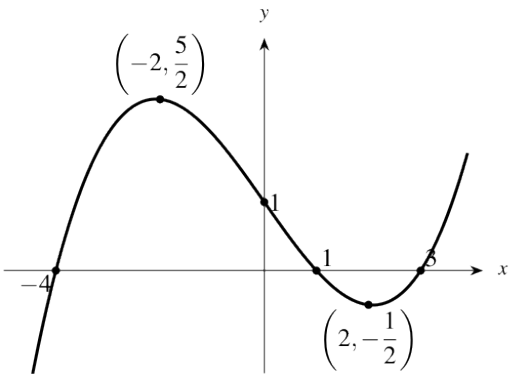
FF75 Further Functions · original Q75 · source p.74

75

No calculator **Haese Review 6A · Q5(b)**

[Maximum mark: 3]

The graph of $y = f(x)$ is given (the same graph as Section 2). On the same set of axes, graph the functions $y = f(x)$ and $y = [f(x)]^2$.



FF74 Further Functions · original Q74 · source p.72

74 *Haese AA HL Ex 6B · Q7*

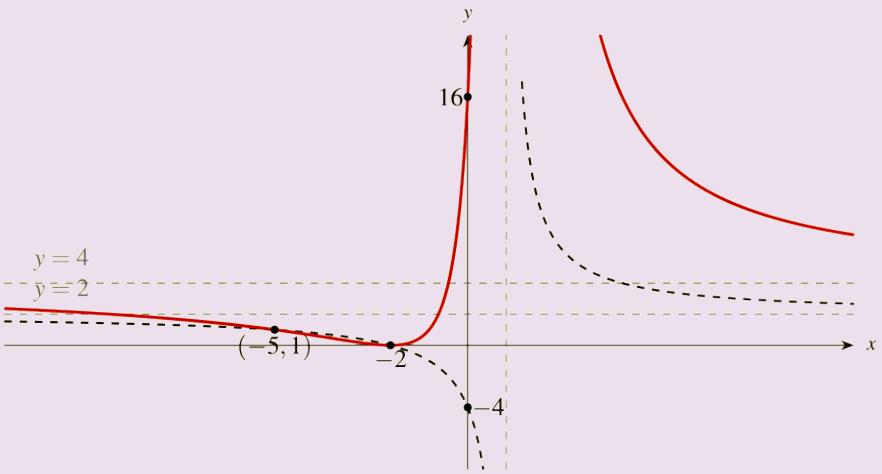
(a) x-int -2, y-int -4, V.A. $x=1$, H.A. $y=2$ **A2**

(b) x-int -2 (touches), y-int 16, V.A. $x=1$, H.A. $y=4$ **A2**

(c) $f(x)=0$: $(-2,0)$; $f(x)=1$: $2x+4=x-1 \Rightarrow x=-5$: $\boxed{(-2,0) \text{ and } (-5,1)}$ **A1**

A1

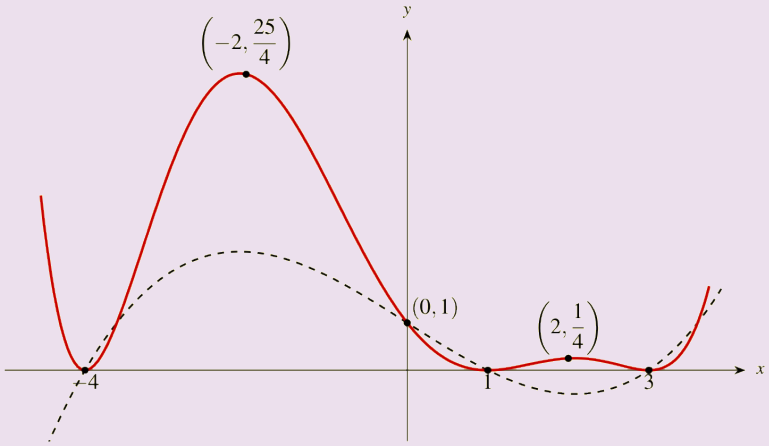
(d) **A2**



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75 *Haese Review 6A · Q5(b)*

(dashed: $y=f(x)$; red: new graph)



Touches the axis at -4, 1, 3; maxima $\left(-2,\frac{25}{4}\right)$ and $\left(2,\frac{1}{4}\right)$; $(0,1)$ is invariant. **A3**

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76

No calculator **Haese Review 6A · Q8**

[Maximum mark: 4]

Suppose $f(x) = e^x - 1$.

(a) On the same set of axes, sketch $y = f(x)$ and $y = [f(x)]^2$. [2]

(b) Find the invariant points when $y = f(x)$ is transformed to $y = [f(x)]^2$. [2]

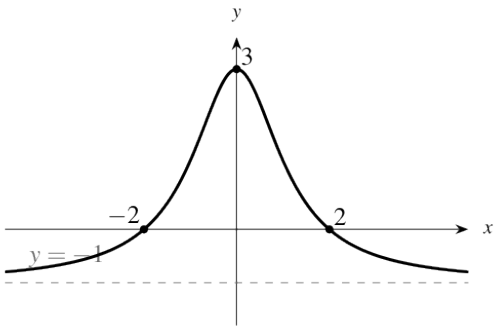
FF77 Further Functions · original Q77 · source p.76

77

No calculator **Haese Review 6B · Q4**

[Maximum mark: 3]

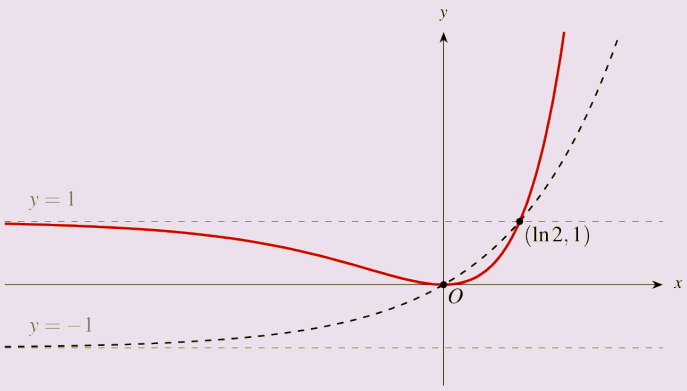
The graph of $y = f(x)$ is given. On the same set of axes, graph the functions $y = f(x)$ and $y = [f(x)]^2$.



FF76 Further Functions · original Q76 · source p.74

76 *Haese Review 6A · Q8*

(a)



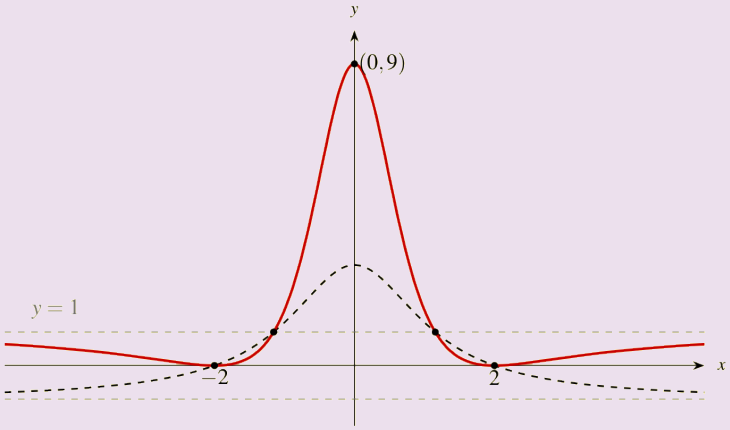
A1 A1

(b) $f(x) = 0 \Rightarrow x = 0$; $f(x) = 1 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$: $\boxed{(0,0) \text{ and } (\ln 2,\ 1)}$ **A1 A1**

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77 *Haese Review 6B · Q4*

(dashed: $y = f(x)$; red: new graph)



Maximum $(0,3) \to (0,9)$; touches at ± 2; H.A. $y = -1$ becomes $y = 1$ (approached from below); invariant where $f = 1$: $x = \pm\dfrac{2}{\sqrt{3}}$. **A3**